



A GUIDE FOR FAMILIES

Quality Time

100+ screen-free activities and experiments for kids ages 0 to 12

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Quality Time — 100+ screen-free activities and experiments for kids ages 0 to 12

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- Experiment
- Sensory
- Active play
- Game
- Connection
- Arts & crafts
- Nature

Fewer screens, more moments

We live surrounded by screens. They're useful — but when they take the place of play, conversation and creative boredom, children lose something precious: the chance to discover the world with their own hands, and to feel truly accompanied. \n\n This book is built on one simple idea: the best "content" for a child is you. You don't need to spend money or be an expert. With things you already have at home — cardboard, water, flour, pencils, a bedsheet — you can create memories no app will ever match. \n\n Inside you'll find more than 100 activities organized by age and by type: experiments, sensory play, active games, homemade board games, connection rituals, art and nature. Each one clearly shows how much prep it needs, what materials you'll use and how long it takes, so you can choose based on the time and energy you have today.

How to use this book

1. Find your child's age chapter. Activities from nearby ages are easy to adapt.
2. Check each activity's icons: they tell you at a glance how much prep it needs, what materials you'll use and how long it takes.
3. No need to go in order. Pick based on the day's time and mood.
4. Read the steps first, and check the safety note whenever it appears.
5. Adapt, repeat and improve. The best activities are the ones that become family traditions.

Before you start: safety

These activities are designed for family fun, but they always require adult supervision. Read this page before you begin, and adapt every activity to your child's age and abilities.

- Always supervise. No activity should be done without an adult present, especially with children under 3.
- Small parts. With babies and toddlers, avoid objects that fit entirely in the mouth: they are a choking hazard.
- Heat and fire. Candles, matches and heat sources are always handled by an adult; keep hair, clothing and paper away.
- Magnets. Use large magnets only, never near the mouth; swallowing them is extremely dangerous. Keep them away from electronics.
- Electricity. Use small batteries only, under supervision; never plugs or mains power.
- Allergies and materials. Check for possible allergies (foods, latex, paints) and choose non-toxic materials.
- Water. Keep water away from outlets, and never leave a child alone near water.
- Cleanup. Wash hands when you finish, especially after the experiments.

Legal notice: the information in this book is provided in good faith, for educational and recreational purposes. The author and publisher accept no liability for damage, injury or loss arising from these activities. By using this book you agree to do so at your own risk and with proper adult supervision.

Icon guide

Every activity shows these at-a-glance indicators:

Prep ahead



Low — ready to start right away



Medium — needs 5 to 15 minutes of prep



High — needs 15+ minutes or planning ahead

Materials



Minimal — 1 to 3 things you already have at home



A few — 4 to 6 simple materials



Several — 7 or more household materials

Duration



Approximate activity time, shown with the clock. Adjust it to your child's pace.

Activity types



Experi-
ment



Sens-
ory



Active
play



Game



Connec-
tion



Arts &
crafts



Nature



Ages 0-2

Discovering the world through the senses

Ages 0–2

Discovering the world through the senses

In the first two years, the brain grows faster than at any other time in life. Babies learn by touching, watching, listening and — above all — by feeling accompanied. They don't need screens: they need your voice, your face and your hands. The activities in this chapter are short, safe and designed to share skin to skin. Follow your child's pace: if they get tired or distracted, take a break. The goal isn't to "finish" — it's to be present.

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ACTIVITIES

8

SCIENCE EXPERIMENTS

18

READY TO START NOW

In this chapter

● Treasure Basket	10-15 min	● Magic Shadows	10-15 min
● Bubble Chasers	10-15 min	● Disappearing Ice	5-10 min
● Calm-Down Bottle	10-15 min	● Ball Ramp	10-15 min
● Splash and Pour	10-20 min	● Catch the Light	5-10 min
● Texture Blanket	10-15 min	● Falling Paper	5-10 min
● Where's Mommy? Peekaboo!	5-10 min	● Backyard Treasures	10-15 min
● Faces in the Mirror	5-10 min	● Plant a Seed	10-15 min
● Little Kitchen Band	10-15 min	● Sponge Stamps	10-15 min
● Pillow Obstacle Course	10-20 min	● Roll It Back and Forth	5-10 min
● Tumbling Towers	10-15 min	● Sing-Along Massage	10-15 min
● Yogurt Finger Paint	10-15 min	● Scarf Dance	10-15 min
● In and Out	10-15 min	● Bubble Bottle	10-15 min
● Sink or Float?	10-15 min	● Rolling on Every Surface	10-15 min

Treasure Basket

A basket full of everyday objects for the baby to explore with all their senses.

PREP

MATERIALS


 10-15
min


Materials

- A basket, shallow box, or tray
- Safe objects with different textures and materials (wooden spoon, sponge, metal cup, fabric scrap, cloth ball)
- A blanket for the floor

Prep ahead

Choose 6 to 10 safe objects from around your house, made of different materials (wood, fabric, metal, silicone). Wash them and make sure none are small enough to fit entirely in the baby's mouth or have parts that could come loose. Place them in the basket.

Step by step

- 1 Sit the baby on the blanket with the basket within reach. If they can't sit up on their own yet, lay them on their side facing the basket.
- 2 Let the baby discover the objects at their own pace. Don't direct the play: just stay close and watch.
- 3 Name whatever they pick up: "the wooden spoon," "ooh, the cup is cold!" Your voice gives them reassurance and vocabulary.
- 4 When they tire of one object, let them choose another. Remove the ones that lose their appeal and offer new ones another day.

What it builds: Stimulates touch, hand-eye coordination, and language. The baby compares textures, weights, and temperatures, building the foundation for scientific thinking and focus.

Variations & tips: Swap out the objects every week to keep things fresh. In summer, add a cold object that has spent a little while in the fridge for a brand-new sensation.

Safety: Supervise at all times. Avoid objects that are small, pointy, have long strings, or could break. When in doubt, leave it out.



We did it!

Date: _____



How did they react? What caught their attention?

Glue a photo of the moment here:

• • •

Bubble Chasers

Soap bubbles that float, shimmer, and vanish at a touch, pure magic in a baby's eyes.

PREP

MATERIALS


 **10-15**
min


Materials

- Warm water in a container
- A squirt of gentle soap (dish soap or hand soap)
- A straw or a wand made from coated wire

Prep ahead

Mix the warm water with a little soap in a container. Do this yourself, out of the baby's reach, and keep the mixture well away from their hands so it doesn't end up in their mouth.

Step by step

- 1 Sit or hold the baby somewhere comfortable, indoors or out in the yard.
- 2 Gently blow bubbles in front of them, without getting too close to their face.
- 3 Narrate what's happening: "look, it's going up!" "pop, it's gone!" Add your voice and your smile.
- 4 If they're already crawling or walking, let them try to touch the bubbles or follow them with their eyes and hands.

What it builds: Develops visual tracking, attention, and the idea of things appearing and disappearing. Bubbles invite movement and those first shared giggles.

Variations & tips: Blow bigger, slower bubbles for younger babies, or lots of little ones for babies who are already on the move and want to chase them.

Safety: Never let the baby drink or blow into the mixture. Keep the soapy liquid out of their reach and away from their eyes. Supervise at all times.



We did it!

Date: _____



How did they react? What caught their attention?

Glue a photo of the moment here:

• • •

Calm-Down Bottle

A bottle filled with water, sparkles, and floating objects that mesmerizes and soothes the baby.

PREP

MATERIALS



10-15
min



Materials

- A small plastic bottle with a cap
- Water
- Oil or a little clear gel (optional)
- Small, eye-catching objects (colorful buttons, sequins, glitter, bottle caps)
- Strong glue or tape to seal the cap

Prep ahead

Fill the bottle with water and add the eye-catching objects. Close the cap tightly and seal it with strong glue or tape so it can't be opened under any circumstances. Check that it's completely sealed before handing it to the baby.

Step by step

- 1 Show the baby the bottle and shake it gently so they can see everything moving inside.
- 2 Roll it slowly in front of them so they follow it with their eyes.
- 3 Hand it over so they can hold it, turn it, and shake it at their own pace.
- 4 Name the colors and the movements: "watch the glitter drift down," "everything's spinning!"

What it builds: Stimulates visual tracking, cause and effect, and concentration. Its slow-moving contents help calm the baby during fussy moments.

Variations & tips: Make several bottles with different fillings: water and oil, colored rice, or little pom-poms, to vary the sights and sounds.

Safety: Sealing the cap is essential: if the bottle ever opened, the small objects would be a choking hazard. Check it before every use and always supervise.



We did it!

Date: _____



How did they react? What caught their attention?

Glue a photo of the moment here:

Splash and Pour

Containers, spoons, and water for splashing, filling, and dumping over and over again.

PREP

MATERIALS


 10-20
min


Materials

- A shallow bucket or basin
- A small amount of warm water
- Cups, mugs, and spoons in different sizes
- A towel for the floor

Prep ahead

Lay a towel on the floor and set the bucket on top with a little warm water. Dress the baby in clothes that can get wet, or just leave them in a diaper. Fill it only a few inches deep.

Step by step

- 1 Sit the baby in front of the bucket, always right beside you and within arm's reach.
- 2 Show them how to fill a cup and pour it back into the water.
- 3 Let them splash, dip their hands in, and move water from one container to another.
- 4 Add words as you go: "full," "empty," "splash!" Enjoy the play without rushing.

What it builds: Builds fine motor skills, coordination, and the concepts of full and empty. Water offers a calming, wonderfully engaging sensory experience.

Variations & tips: Add floating toys or sponges to squeeze. In summer, take it outside so they can get as wet as they like.

Safety: Never leave the baby alone near water, not even for a second: just a couple of inches is enough to pose a risk. Use very little water and dry the floor to prevent slips.



We did it!

Date: _____



How did they react? What caught their attention?

Glue a photo of the moment here:

• • •

Texture Blanket

A blanket or panel covered in different fabric scraps to touch, rub, and discover.

PREP

MATERIALS



10-15
min



Materials

- Scraps of different fabrics (terry cloth, felt, satin, corduroy, wool)
- A large base fabric or a thin blanket
- Needle and thread or fabric glue

Prep ahead

Sew or glue several scraps of different textures firmly onto the base fabric, well attached so they can't peel off. Check that there are no loose edges or long threads. This prep takes a while, but the blanket will last a long time.

Step by step

- 1 Spread the texture blanket on the floor and lay or sit the baby on top of it.
- 2 Guide their hand across the different fabrics and let them touch freely.
- 3 Describe what they're feeling: "soft," "rough," "squishy." Repeat the words calmly.
- 4 Notice which texture they like best and return to it as often as they want.

What it builds: Refines the sense of touch and enriches sensory vocabulary. Comparing textures helps the baby's brain organize information about the world.

Variations & tips: Add a patch with a piece of crinkly plastic or a little bell sewn inside to bring sound into the experience.

Safety: Make sure every scrap is firmly attached, with no small pieces that could come loose. Avoid loose sequins or buttons. Always supervise.



We did it!

Date: _____



How did they react? What caught their attention?

Glue a photo of the moment here:

Where's Mommy? Peekaboo!

The classic hide-and-appear game that teaches that what's out of sight is still there.

PREP

MATERIALS



5-10

min



Materials

- A scarf, a thin cloth, or just your own hands

Prep ahead

No prep needed. Just pick a moment when the baby is calm and alert, and position yourself in front of them at eye level.

Step by step

- 1 Cover your face with your hands or a scarf and ask: "where's Mommy?"
- 2 Wait a beat, then reveal your face all at once, saying "peekaboo!" with a big smile.
- 3 Repeat the game at your own pace, stretching out the pause to build anticipation.
- 4 Once they know the game, drape the cloth over the baby and let them be the one who pops out.

What it builds: Teaches object permanence: even when something disappears, it still exists. Strengthens bonding, trust, and emotional regulation around separation.

Variations & tips: Hide a toy under the cloth and ask where it went, so they can search for it and uncover it themselves.

Safety: Use thin fabrics and never fully cover the baby's face. Stay right in front of them and watch to make sure the cloth doesn't cover their nose or mouth.



We did it!

Date: _____



How did they react? What caught their attention?

Glue a photo of the moment here:

Faces in the Mirror

Together at the mirror, discovering smiles, silly faces, and their very own reflection.

PREP

MATERIALS



5-10

min



Materials

- A safe mirror (wall-mounted or shatter-proof)

Prep ahead

Choose a mirror that's securely mounted to the wall or a shatterproof one with no sharp edges. Position yourself with the baby at a comfortable distance where you can both see yourselves clearly.

Step by step

- 1 Hold the baby in front of the mirror and let them study the reflections.
- 2 Point and name: "that's the baby," "that's Mommy," gently touching their reflection.
- 3 Make happy faces, surprised faces, or blow a kiss, and watch how they imitate or react.
- 4 Put the emotions into words: "so happy!" "what a surprise!"

What it builds: Supports self-recognition, emotional connection, and imitation, the foundation of empathy and social learning.

Variations & tips: Move a toy behind the baby so it appears in the mirror, and let them discover where it's coming from when they turn around.

Safety: Check that the mirror is firmly secured and can't fall or break. Avoid glass mirrors within the baby's reach without supervision.



We did it!

Date: _____



How did they react? What caught their attention?

Glue a photo of the moment here:

Little Kitchen Band

Pots, spoons, and jars turn into drums and maracas for making music together.

PREP

MATERIALS



10-15
min



Materials

- Pots, pans, or plastic containers
- Wooden spoons
- A tightly sealed jar with a little rice or dried beans inside (maraca)

Prep ahead

Gather a few kitchen utensils and a homemade maraca made from a well-sealed jar. Make sure the maraca can't be opened so the grains stay inside.

Step by step

- 1 Sit the baby on the floor and arrange the utensils around them.
- 2 Show them how to gently tap a pot with the spoon and shake the maraca.
- 3 Sing a simple song and keep the beat together.
- 4 Let them experiment with the different sounds and volumes in their own way.

What it builds: Develops a sense of rhythm, coordination, and cause and effect. Making music together stimulates language and family fun.

Variations & tips: Alternate loud and soft moments, fast and slow, so they discover the contrast between sounds.

Safety: Check that the maraca is perfectly sealed to prevent a choking hazard from the grains. Use utensils without sharp edges and always supervise.



We did it!

Date: _____



How did they react? What caught their attention?

Glue a photo of the moment here:

• • •

Pillow Obstacle Course

A trail of pillows and blankets for crawling, climbing, and rolling without any danger.

PREP

MATERIALS


 10-20
min


Materials

- Cushions and pillows in different sizes
- Blankets or comforters
- Open cardboard boxes (optional)

Prep ahead

Clear a soft area of the floor and build little mountains, ramps, and a tunnel with cushions, blankets, and, if you like, a cardboard box. Keep everything away from furniture with hard corners.

Step by step

- 1 Place the baby at the start of the course and encourage them to make their way toward you.
- 2 Help them climb over a cushion or crawl under a blanket tunnel.
- 3 Celebrate every milestone: "way to go!" "you made it!"
- 4 Let them repeat and explore the course at their own pace, switching up the obstacles.

What it builds: Stimulates gross motor skills, balance, and confidence in their own body. Conquering little challenges builds self-esteem.

Variations & tips: Hide a toy at the end of the course as the finish line, or lie down at the other end yourself so they crawl all the way to you.

Safety: Pad the whole area well and move furniture, corners, and hard objects out of the way. Don't leave cushions that could cover a young baby's face, and always supervise.



We did it!

Date: _____



How did they react? What caught their attention?

Glue a photo of the moment here:

• • •

Tumbling Towers

Stack cups or boxes into a tower, then knock it all down with a single swipe.

PREP

MATERIALS



10-15
min



Materials

- Plastic cups, plastic containers, or small lightweight boxes

Prep ahead

Gather several lightweight cups or little boxes that stack. Check that they have no sharp edges or loose parts and that they're big enough not to fit in the baby's mouth.

Step by step

- 1 Sit the baby across from you with the cups between the two of you.
- 2 Stack two or three cups into a tower and let them take a good look.
- 3 Encourage them to knock it down with their hand: "boom, it fell!"
- 4 Build it again and let them try stacking or toppling it themselves, over and over.

What it builds: Works fine motor skills, cause and effect, and the idea of balance. Building and knocking down teaches patience and delivers fun that never gets old.

Variations & tips: As they grow, offer cups in different sizes so they discover how they nest inside one another.

Safety: Use lightweight, unbreakable materials so nothing hurts them when it falls. Avoid small pieces and always supervise.



We did it!

Date: _____



How did they react? What caught their attention?

Glue a photo of the moment here:

Yogurt Finger Paint

Finger painting with colorful yogurt, a work of art that's perfectly safe to taste.

PREP

MATERIALS



10-15
min



Materials

- Plain yogurt
- A little food coloring or fruit puree for color
- Thick paper or a tray
- A towel or large bib

Prep ahead

Divide the yogurt into small bowls and tint it with a little food coloring or fruit puree. Cover the table or floor, dress the baby in clothes that can get messy, and keep a towel handy.

Step by step

- 1 Sit the baby in front of the paper or tray with the bowls of colored yogurt.
- 2 Dip your own finger first and make a mark to show them the idea.
- 3 Let them touch, smear, and mix the colors freely with their hands.
- 4 Stay close without correcting: the goal is to explore, not to make a pretty picture.

What it builds: Stimulates touch, creativity, and coordination. Because it's edible, it lets them explore with all their senses, worry-free.

Variations & tips: Use two or three colors and let them discover what happens when they mix. You can also paint right on the high chair tray.

Safety: Use only edible ingredients and food coloring, since everything will end up in their mouth. Check for yogurt allergies first and always supervise.



We did it!

Date: _____



How did they react? What caught their attention?

Glue a photo of the moment here:

• • •

In and Out

A box and a handful of objects for filling, emptying, and discovering the joy of things appearing and disappearing.

PREP

MATERIALS



10-15
min



Materials

- A cardboard box or a wide-mouthed container
- Cloth balls, soft toys, or rolled-up socks
- A blanket for the floor

Prep ahead

Choose a box or container the baby can easily reach into and gather several soft, safe objects, big enough that they can't fit entirely in the baby's mouth. Set everything out on the blanket.

Step by step

- 1 Sit the baby on the blanket with the box and objects within reach.
- 2 Show them how to put an object into the box and take it back out, naming "in" and "out."
- 3 Encourage them to copy you and let them fill and empty the box at their own pace, again and again.
- 4 Celebrate every success: "you put it in!" "and now you took it out!"

What it builds: Works fine motor skills, hand-eye coordination, and the concepts of in and out. Filling and emptying strengthens concentration and the idea of object permanence.

Variations & tips: Offer containers in different sizes, or a box with a smaller opening to add a challenge once they've mastered the game.

Safety: Use only soft, large objects with no parts that could come loose, and keep plastic bags out of reach. Supervise at all times to prevent choking.



We did it!

Date: _____



How did they react? What caught their attention?

Glue a photo of the moment here:

• • •

EXPERIMENT

Ages 0–2

Sink or Float?

A bucket of water and household objects to discover which ones stay on top and which drop to the bottom.

PREP

MATERIALS



10-15
min



Materials

- A large clear bucket or container with water
- Safe, large objects that float (a sponge, a plastic ball, a plastic pot lid)
- Safe, large objects that sink (a metal spoon, a small ceramic cup)
- A towel for the floor

Prep ahead

Fill the bucket halfway with warm water and set it on a towel on the floor. Gather a few large, safe, freshly washed objects that can't fit entirely in the baby's mouth.

Step by step

- 1 Sit the baby in front of the bucket, somewhere it's fine to get wet.
- 2 Take an object and drop it into the water while you say "look, it floats!" or "it sank!"
- 3 Hand the baby an object and let them be the one to drop it and watch what happens.
- 4 Name what happens every time and celebrate their surprise: "the spoon went all the way down!"
- 5 Let them repeat with the same objects as many times as they like; repetition is part of the learning.

What it builds: Introduces cause and effect and first notions of weight and materials. The baby predicts, tests, and marvels, the foundation of scientific thinking.

Variations & tips: Add a cold object straight from the fridge to bring temperature into the mix, or use just one type of object so they notice the pattern.

Safety: Never leave the baby alone near water, not even for an instant. Use only large objects to prevent choking and keep the floor around you dry so nobody slips.

 **We did it!**

Date: _____ ☆ ☆ ☆ ☆ ☆

How did they react? What caught their attention?

Glue a photo of the moment here:

 EXPERIMENT

Ages 0–2

Magic Shadows

With a flashlight and a wall, hands and toys come alive as shadows that appear and grow.

PREP

MATERIALS



10-15
min



Materials

- A flashlight or a phone's flashlight
- A light-colored wall or sheet
- A few toys with recognizable shapes (a stuffed animal, a spoon, a ball)

Prep ahead

Pick a room you can dim, but not make fully dark, so the baby doesn't get scared. Point the flashlight at a light-colored wall or a hanging sheet.

Step by step

- 1 Sit or hold the baby facing the wall, with you right beside them.
- 2 Turn on the flashlight and move your hand between the light and the wall to make a shadow.
- 3 Bring your hand closer and farther so the shadow grows and shrinks, and narrate what's happening.
- 4 Hold a toy in front of the light and ask, "what could that shadow be?"
- 5 Let the baby touch the wall where the shadow is and discover there's nothing to grab.

What it builds: Explores the relationship between light, object, and shadow, a first experience of cause and effect and of how distance changes size. Stimulates visual attention and curiosity.

Variations & tips: Make simple animal shadows with your hands, or let a baby who's already walking move their own body to watch their shadow dance.

Safety: Don't shine the light directly into the baby's eyes. Keep a soft background light on so the dimness doesn't frighten them, and stay with them the whole time.



We did it!

Date: _____



How did they react? What caught their attention?

Glue a photo of the moment here:

• • •

 EXPERIMENT

Ages 0–2

Disappearing Ice

A big ice cube that turns to water in the baby's warm hands, cold that slips away.

PREP

MATERIALS

● ○ ○

■ □ □



5-10

min



Materials

- A large ice cube (from a big mold, never a small one)
- A tray or shallow bowl
- A small towel

Prep ahead

Make a large ice cube ahead of time using a wide mold or a cup, so it's far too big to fit whole into the baby's mouth. Have it ready in the freezer.

Step by step

- 1 Sit the baby down and place the tray with the ice within reach, with the towel nearby.
- 2 Touch the ice first yourself and say "brrr, so cold!" with a little gesture of surprise.
- 3 Guide the baby's hand to touch the ice for just a moment and watch their reaction.
- 4 Show them how the ice leaves water behind and gets smaller as time passes.
- 5 Dry their hands with the towel if they feel too cold, and let them watch it melt.

What it builds: Discovers cold, a brand-new temperature, and witnesses a change of state: solid becoming liquid. Works touch, attention, and the idea of transformation.

Variations & tips: Freeze a leaf or a large, visible piece of fruit inside the ice so the baby can watch it appear as the ice melts.

Safety: Don't let the ice stay in contact with skin for more than a few seconds to avoid cold burns. Always use large ice to prevent choking and supervise closely.



We did it!

Date: _____



How did they react? What caught their attention?

Glue a photo of the moment here:

• • •

EXPERIMENT

Ages 0–2

Ball Ramp

A tilted board and balls that roll down all by themselves: gravity at play right before the baby's eyes.

PREP

MATERIALS



10-15
min



Materials

- A board, sturdy tray, or thick cardboard to make the ramp
- A cushion or low box to prop up and tilt the ramp
- Large, safe balls or large toy cars

Prep ahead

Rest one end of the board on a cushion or low box to form a gentle ramp. Check that it's stable and that the balls are large, with no small parts.

Step by step

- 1 Sit the baby at the bottom of the ramp, where the balls will arrive.
- 2 Place a ball at the top and let it go, saying "ready? Here it comes!"
- 3 Let the baby watch it roll down on its own and catch it or chase it at the bottom.
- 4 Hand them a ball so they can place it at the top themselves and discover that it rolls too.
- 5 Switch objects: try a ball and then a car, and see which one gets down faster.

What it builds: Experiences gravity and motion: objects roll down a slope all on their own. Works anticipation, visual tracking, and the coordination it takes to catch.

Variations & tips: Change the ramp's tilt so the balls roll slower or faster, or put a towel at the bottom so they come to a gentle stop.

Safety: Use only large balls and objects to prevent choking. Secure the ramp well so it can't tip over onto the baby, and stay with them throughout the game.



We did it!

Date: _____



How did they react? What caught their attention?

Glue a photo of the moment here:

• • •

 EXPERIMENT

Ages 0–2

Catch the Light

A little mirror catches the sun and makes a spot of light dance across the wall for the baby to chase.

PREP

MATERIALS



5-10

min



Materials

- A small hand mirror or a shiny metal object
- Sunlight coming through a window
- A light-colored wall or floor

Prep ahead

Find a time of day when the sun comes through a window. Position yourself so you can bounce the light with the mirror onto a wall or the floor near the baby.

Step by step

- 1 Sit the baby somewhere comfortable facing the sunlit wall.
- 2 Tilt the mirror until a spot of light appears on the wall.
- 3 Move the mirror gently so the light travels, and narrate: "there goes the light!"
- 4 Let the baby follow the reflection with their eyes and, if they crawl, try to reach it.
- 5 Hold the light still near their hand so they feel like they almost caught it.

What it builds: Discovers how light reflects and the cause-and-effect link between your movement and the glowing spot. Stimulates visual tracking, attention, and purposeful movement.

Variations & tips: Try different shiny objects, like a metal spoon, and compare the reflections each one makes.

Safety: Never aim the reflection at the baby's eyes. Keep the mirror in your own hands and, if it's glass, don't hand it over; supervise at all times.



We did it!

Date: _____



How did they react? What caught their attention?

Glue a photo of the moment here:

• • •

EXPERIMENT

Ages 0–2

Falling Paper

A sheet of paper drifts down slowly while a ball drops in a flash: two very different ways of falling that surprise the baby.

PREP

MATERIALS

5-10
min

Materials

- Sheets of paper or tissues
- A light, large object that falls fast (a cloth ball)
- A clear, open space

Prep ahead

Gather a few sheets of paper and one large, safe object. All you need beyond that is an open space where the baby can comfortably look up.

Step by step

- 1 Sit or lay the baby facing the spot where you'll drop things.
- 2 Release a sheet of paper from up high and say, "look how slowly it floats down!"
- 3 Drop the cloth ball and compare: "that one fell fast!"
- 4 Repeat, alternating paper and ball, so they notice the difference.
- 5 Hand them a sheet of paper to let go themselves and watch it float to the floor.

What it builds: Compares how light and heavy objects fall, a first taste of gravity and air. Works observation, anticipation, and downward visual tracking.

Variations & tips: Crumple the paper into a little ball and drop it: the baby will see that the very same paper now falls faster than when it was flat.

Safety: Watch that the baby doesn't put the paper in their mouth, since it can be a choking hazard. Pick up any small pieces and stay with them throughout the game.



We did it!

Date: _____



How did they react? What caught their attention?

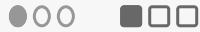
Glue a photo of the moment here:

• • •

Backyard Treasures

A slow stroll through the grass to touch leaves, feel the breeze, and discover the natural world with every sense.

PREP MATERIALS



🕒 10-15 min



Materials

- A blanket for the ground
- A backyard, a park, or a flower pot with plants
- Comfortable clothes for the weather

Prep ahead

Choose a safe outdoor spot in the shade. Check the area first and clear away small stones, litter, or plants the baby might put in their mouth.

Step by step

- 1 Spread the blanket on the grass and sit or lay the baby on top.
- 2 Carefully bring a big leaf or a flower close so they can touch it and look at it.
- 3 Name what they discover: "this is a leaf, it's soft," "listen to the wind!"
- 4 Let them feel the grass under their bare hands or feet if the weather allows.
- 5 Stay by their side, watching the birds and taking in the sounds of the place together.

What it builds: Connects the baby with nature and multiplies their sensory experiences: textures, smells, sounds, and temperature. Encourages calm, curiosity, and attention.

Variations & tips: Gather a few large, safe leaves in a basket yourself so the baby can explore them sitting on the blanket without wandering off.

Safety: Watch that they don't put dirt, leaves, flowers, or small stones in their mouth, since they could choke or the plants could be toxic. Use sunscreen and shade.



We did it!

Date: _____



How did they react? What caught their attention?

Glue a photo of the moment here:

• • •

Plant a Seed

Fill a flower pot with soil, tuck in a seed, and wait together for the plant to wake up.

PREP

MATERIALS



10-15
min



Materials

- A small flower pot or a large cup
- Potting soil
- A large, fast-growing seed (a bean or an avocado pit)
- A little water in a small bucket

Prep ahead

Have the soil, pot, and seed ready on a tray on an easy-to-clean surface. Work outdoors or somewhere a little mess won't be a problem.

Step by step

- 1 Sit the baby with you and let them touch the soil with their hands while you fill the pot.
- 2 Guide their hand to make a small hole and place the seed inside.
- 3 Cover it with a little soil together, naming what you're doing.
- 4 Help them pour water carefully, then set the pot next to a sunny window.
- 5 Come back every day to look at it and water it, and watch it sprout over time.

What it builds: Introduces the idea that plants are born and grow, along with the notions of care and waiting. Contact with soil is a rich sensory experience and a bond with nature.

Variations & tips: Plant two seeds and put one by the window and the other in a dark corner, then watch together over time to see which grows better.

Safety: Watch that the baby doesn't put soil or the seed in their mouth, since they could choke. Wash their hands well afterward and supervise closely the whole time.

 We did it!

Date: _____ ☆ ☆ ☆ ☆ ☆

How did they react? What caught their attention?

Glue a photo of the moment here:

Sponge Stamps

Sponges dipped in paint leave colorful prints on the paper with just a squeeze.

PREP

MATERIALS



🕒 10-15
min



Materials

- Large sponges cut into big chunks
- Non-toxic finger paint
- Large sheets of paper
- A plate and a tablecloth or newspaper to cover the table

Prep ahead

Cut the sponges into big pieces, easy to grip and too large to fit in the baby's mouth. Pour a little paint onto a plate and cover the table or floor with newspaper.

Step by step

- 1 Sit the baby, dressed in clothes that can get messy, in front of the paper.
- 2 Dip a sponge in the paint and stamp it yourself first to show the print.
- 3 Give the baby a sponge and guide their hand to press it onto the paper.
- 4 Name the colors and celebrate every mark: "look, red!" "another print!"
- 5 Let them stamp freely as many times as they like, switching colors.

What it builds: Stimulates creativity, touch, and hand-eye coordination. The baby discovers cause and effect: squeeze the sponge and a splash of color appears.

Variations & tips: Use sponges in different shapes to compare the prints, or tape paper to a wall so they can stamp standing up.

Safety: Use only non-toxic paint and large sponges to prevent choking. Watch that they don't put the sponges in their mouth and supervise the whole time.



We did it!

Date: _____



How did they react? What caught their attention?

Glue a photo of the moment here:

• • •



GAME

Ages 0–2

Roll It Back and Forth

A ball that rolls from your hands to the baby's and back again, the very first game of taking turns and sharing.

PREP

MATERIALS

5-10
min

Materials

- A soft, large ball
- A clear spot in the living room or on the floor

Prep ahead

Clear a spot on the floor and sit facing the baby with your legs spread wide so the ball can't get away. That's all the prep you need.

Step by step

- 1 Sit facing the baby, just a short distance apart, with the ball in your hands.
- 2 Roll it gently toward them, saying "here it comes, for you!"
- 3 Encourage them to send it back by pushing it with their hands: "now roll it to me!"
- 4 Celebrate every time the ball goes back and forth, even if the push goes sideways.
- 5 Repeat the turn-taking rhythm as many times as their interest lasts.

What it builds: Teaches turn-taking, give-and-take, and waiting, the first social building blocks. Works coordination through pushing and catching, and deepens your bond through shared play.

Variations & tips: Once they're crawling or walking, move a little farther away so they have to go get the ball and bring it back.

Safety: Use a soft, large ball to avoid choking hazards and bumps. Play in a space free of hard objects and stay with them the whole time.



We did it!

Date: _____



How did they react? What caught their attention?

Glue a photo of the moment here:

• • •

♡ CONNECTION

Ages 0–2

Sing-Along Massage

Gentle strokes in time with a song so the baby relaxes and feels your love right on their skin.

PREP

MATERIALS



10-15
min



Materials

- A soft blanket or towel
- A little baby-safe oil or lotion (optional)
- A warm, quiet spot

Prep ahead

Pick a moment when the baby is calm and awake, neither hungry nor sleepy. Warm up the room, wash your hands, and warm a little lotion between your palms.

Step by step

- 1 Lay the baby on their back on the blanket, facing you.
- 2 Start with the legs, using slow, gentle strokes, always keeping eye contact.
- 3 Pair each movement with a quiet song or your soft voice.
- 4 Move on to the arms, hands, and tummy with delicate strokes.
- 5 Stop if the baby turns their face away or gets fussy; respect their signals.

What it builds: Strengthens bonding, trust, and emotional regulation through touch. Massage promotes calm, better sleep, and body awareness.

Variations & tips: Use the same song every time to build a routine the baby comes to associate with winding down before sleep.

Safety: Test the lotion on a small patch of skin first in case of allergies, and keep it away from the eyes. Don't massage if the baby is sick or has just eaten, and never leave them alone on a high surface.



We did it!

Date: _____



How did they react? What caught their attention?

Glue a photo of the moment here:

• • •

Scarf Dance

Featherlight scarves that float and swirl through the air while you dance together to the music.

PREP

MATERIALS



10-15
min



Materials

- Scarves or large, lightweight fabrics
- Soft, cheerful music
- A clear space in the living room

Prep ahead

Clear a space so you can move around safely and pick some calm, cheerful music. Keep a couple of large, lightweight scarves within reach.

Step by step

- 1 Hold the baby in your arms or sit them in front of you and start the music.
- 2 Wave a scarf through the air so it floats and falls, naming "up!" and "down!"
- 3 Brush it gently over their arms and cheeks so they feel the fabric.
- 4 Dance, swaying to the beat, raising and lowering the scarf with the music.
- 5 Hand them a scarf so they can shake it and wave it their own way.

What it builds: Stimulates rhythm, movement, and coordination, plus visual tracking of the scarf in the air. Dancing together strengthens bonding and joy.

Variations & tips: Alternate fast and slow songs so the baby notices the change of rhythm in your movements and in the scarf's flight.

Safety: Never let the scarf cover the baby's face or wrap around their neck. Use large, lightweight fabrics and supervise the whole time.



We did it!

Date: _____



How did they react? What caught their attention?

Glue a photo of the moment here:

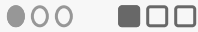
EXPERIMENT

Ages 0–2

Bubble Bottle

A bottle of water with a little air inside that fills with bubbles when you shake it, all rising to the top on their own.

PREP MATERIALS


 10-15
min


Materials

- A small clear plastic bottle with a tight-fitting cap
- Tap water
- A squirt of dish soap or gentle shampoo
- Strong tape to seal the cap

Prep ahead

Fill the bottle with water to about three finger-widths from the top, add a small squirt of soap, and leave some air inside. Close the cap very tightly and seal around it with tape so it can't open or leak.

Step by step

- 1 Sit the baby with you and show them the bottle, still and clear.
- 2 Shake it slowly and let them watch lots of bubbles appear inside the water.
- 3 Stop and watch together as the bubbles rise to the top on their own and disappear.
- 4 Give the baby the bottle so they can shake it and discover that they can make bubbles too.
- 5 Add words as you go: "so many bubbles!" "look, they're going up," "now they're gone."

What it builds: Discovers cause and effect: shaking the bottle makes the air and soap form bubbles that then rise on their own. Works visual tracking, attention, and the idea that their actions make things happen.

Variations & tips: Make another bottle with warm tap water so they notice the temperature, or add a few drops of food coloring so the bubbles show up in color.

Safety: Seal the cap very well with tape so the baby can't open it or drink the soapy water. Use a large bottle with no small parts that could come loose to prevent choking, and stay with them throughout the game.



We did it!

Date: _____



How did they react? What caught their attention?

Glue a photo of the moment here:

• • •

EXPERIMENT

Ages 0–2

Rolling on Every Surface

The same ball with the same push: on a towel it barely moves, but on the smooth floor it goes flying.

PREP

MATERIALS



🕒 10-15
min



Materials

- A large, safe ball that rolls well
- A towel or blanket
- A rug or a piece of thick fabric
- A stretch of smooth floor, like tile or wood

Prep ahead

Clear a space on the floor and keep the large ball handy. Lay out the towel and the rug nearby so you have three different surfaces: smooth floor, towel, and rug.

Step by step

- 1 Sit the baby facing you on the smooth floor, a couple of steps away.
- 2 Push the ball toward them and watch together how far and fast it rolls.
- 3 Repeat the same push on the towel and show them how much less the ball travels.
- 4 Now try it on the rug and name what happens: "it rolls slowly here."
- 5 Let the baby push the ball on each surface and discover the difference.
- 6 Celebrate their surprise and try again as many times as they want.

What it builds: Experiences how the surface changes the motion: on smooth ground the ball rolls easily, and on rough ground it slows down. Works cause and effect, comparison, and the coordination of pushing and catching.

Variations & tips: Add a fourth surface, like a sturdy tray, or use a smaller ball, still at least as big as their fist, to compare how it rolls on each surface.

Safety: Use only large balls with no small parts that could come loose, to prevent choking. Clear the space of anything they could bump into and stay with them throughout the game.



We did it!

Date: _____



How did they react? What caught their attention?

Glue a photo of the moment here:

• • •



Ages 3-5

Play is learning

Ages 3–5

Play is learning

At this age imagination takes off and everything becomes a game. They want to touch, mix, build and ask "why?" nonstop. Channel that curiosity with simple experiments, pretend play and sensory activities. Give them time to explore their own way, and celebrate attempts more than results. A cardboard box can be a spaceship — all you need to do is join the adventure.

26

ACTIVITIES

11

SCIENCE EXPERIMENTS

14

READY TO START NOW

In this chapter

● The Baking Soda Volcano	20-30 min	● White Flowers That Change Color	20-30 min
● The Dancing Colors	20-30 min	● The Candle Under the Glass	20-30 min
● Homemade Squishy Dough	20-30 min	● The Balloon That Blows Itself Up	20-30 min
● The Mystery Box	20-30 min	● Seeds Sprouting in Cotton	20-30 min
● The Pillow Obstacle Course	20-30 min	● The Magnet and Its Hidden Treasures	20-30 min
● Bottle Bowling	20-30 min	● A Rainbow from Water and a Mirror	20-30 min
● The Pretend Store	20-30 min	● Nature Treasure Walk	20-30 min
● The Color Hunt	20-30 min	● Fluffy Foam Painting	20-30 min
● The Faces of Feelings	20-30 min	● The At-Home Treasure Hunt	20-30 min
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● Leaf and Flower Collage	20-30 min	● Bubble-Print Paintings	20-30 min
● Backyard Bug Safari	20-30 min	● Magic Rainbow Milk	20-30 min
● Oil and Water Won't Mix	20-30 min	● The Pepper That Runs Away	20-30 min

 EXPERIMENT

Ages 3–5

The Baking Soda Volcano

A homemade volcano that erupts with colorful foam right before your little one's eyes.

PREP

MATERIALS


 20-30
min


Materials

- A cup or small bottle
- Baking soda (2 or 3 tablespoons)
- Vinegar (half a cup)
- Food coloring or a few drops of beet juice
- A deep tray or large plate to catch the overflow

Prep ahead

Place the cup in the center of the tray to catch the foam. Have the baking soda, vinegar, and food coloring within reach. If you like, shape a mountain around the cup with play dough or crumpled aluminum foil so it looks like a real volcano.

Step by step

- 1 Ask your little one to put two or three tablespoons of baking soda into the cup.
- 2 Together, add a few drops of food coloring or juice on top of the baking soda.
- 3 Count out loud, 'three, two, one,' and let your little one pour in the vinegar all at once.
- 4 Watch the bubbly foam rise up and spill over like lava.
- 5 Once it settles, offer more vinegar to repeat the eruption as many times as you like.

What it builds: Introduces the idea that mixing two things together can make something new happen. It builds observation skills, anticipation, and science vocabulary in a fun way.

Variations & tips: Try warm and cold vinegar to see if the foam changes. Use several cups with different colors to set off multiple eruptions at once.

Safety: Vinegar can sting if it gets in the eyes: supervise the mixing and keep your little one from touching their face. It's not for drinking. Do this activity on an easy-to-clean surface.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

 EXPERIMENT

Ages 3–5

The Dancing Colors

Drops of food coloring that chase and swirl into each other in a plate of milk, like magic.

PREP

MATERIALS



20-30
min



Materials

- A shallow bowl or plate
- Milk (enough to cover the bottom)
- Food coloring or juices in several colors
- A drop of dish soap
- A toothpick, cotton swab, or spoon

Prep ahead

Pour milk into the plate until the bottom is well covered and let it sit for a moment so it's completely still. Keep the food coloring and soap nearby. Set the plate on the table, at your little one's height.

Step by step

- 1 Take turns dropping a few drops of each color onto the milk, spaced apart from each other.
- 2 Watch how the drops float there without mixing.
- 3 Dip the tip of the toothpick into a drop of soap.
- 4 Touch the center of the milk with the toothpick and watch the colors shoot away and dance.
- 5 Touch other spots to create new swirls of color.

What it builds: Shows in a visual way that soap changes how the milk behaves, creating movement. It builds observation skills, patience, and a sense of wonder about what's happening.

Variations & tips: Try it with whole milk and with water to compare what happens in each. Use just two colors and see what new color appears where they meet.

Safety: The soap and the colored milk are not for drinking. Supervise the toothpick so your little one doesn't put it in their mouth or poke themselves.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

Homemade Squishy Dough

A soft dough to knead, squish, and shape into a thousand things with their very own hands.

PREP

MATERIALS


 20-30
min


Materials

- Flour (2 cups)
- Fine salt (half a cup)
- Warm water (a little at a time)
- A tablespoon of oil
- Food coloring or juice (optional)
- A large bowl

Prep ahead

Mix the flour and salt in the bowl. Have the warm water and oil within reach. Clear the table and, if you like, cover it with a tablecloth so you can knead freely.

Step by step

- 1 Ask your little one to mix the flour and salt in the bowl with their hands.
- 2 Add the oil and, little by little, a splash of water while you knead together.
- 3 Keep adding water a bit at a time until the dough stops sticking and feels nice and soft.
- 4 If you want color, make a little well, add a few drops of food coloring, and knead until it's spread through.
- 5 Now it's playtime: make little balls, snakes, animals, or squash it flat with your palms.

What it builds: Works the sense of touch and hand strength, the foundation for holding a pencil later on. Kneading calms, focuses, and lets kids create without rules, valuing the process over the result.

Variations & tips: Store the dough in a sealed container to use on other days. Hide buttons or little paper balls inside and hunt for them like treasures.

Safety: It has a lot of salt, so it's not for eating. Supervise the littlest ones so they don't put it in their mouths, and wash hands when you're done.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

The Mystery Box

Reach in without peeking and guess what's hiding inside using only your sense of touch.

PREP

MATERIALS



20-30
min



Materials

- A cardboard box with a lid or a fabric bag
- Scissors (for the grown-up)
- Everyday objects with different shapes and textures (spoon, sponge, ball, sock, clothespin)
- A sock or cloth to cover the hole (optional)

Prep ahead

The grown-up cuts a hole in one side of the box, just big enough for a hand to fit through. Gather several safe, familiar objects. Put the objects inside without letting your little one see.

Step by step

- 1 Ask your little one to reach through the hole without looking inside.
- 2 Encourage them to touch an object and describe what they feel: 'it's soft,' 'it's pointy,' 'it's round.'
- 3 Have them guess what it is before pulling it out to see if they got it right.
- 4 Celebrate the right answers and the tries alike: what matters is exploring with their hands.
- 5 Switch roles and let your little one hide objects for the grown-up to guess.

What it builds: Sharpens the sense of touch and the ability to describe things with words. It exercises memory, imagination, and language by naming what they feel without seeing it.

Variations & tips: Group objects by theme: only kitchen things, only toys, only soft things. Put in two similar objects and have them tell which is which by touch alone.

Safety: The grown-up uses the scissors and keeps small objects out of reach between turns. Avoid pointy objects or anything that could hurt a hand inside the box.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

The Pillow Obstacle Course

A course of jumps, crawls, and balancing acts built from living room pillows and blankets.

PREP

MATERIALS


 20-30
min


Materials

- Cushions and pillows
- A blanket or sheet
- Chairs (to crawl under)
- Tape or a rope to mark the floor (optional)

Prep ahead

Clear a spot in the living room. Gather pillows, blankets, and a couple of chairs. Move away anything with hard corners or that could break near the course.

Step by step

- 1 Lay pillows on the floor like stepping stones to jump from one to the next without touching the floor.
- 2 Set up two chairs with a blanket over them to make a tunnel to crawl through.
- 3 Mark a straight line with tape or rope to walk along like a balance beam.
- 4 Make up the order of the course together and show your little one how to go through it slowly.
- 5 Have them repeat it several times, first slow and then a little faster, at their own pace.

What it builds: Builds balance, coordination, and whole-body strength. Following a course in order works memory and the ability to plan movements.

Variations & tips: Time each run and celebrate every lap. Do the course like animals: hop like a frog, crawl like a cat, tiptoe like a ballerina.

Safety: Make sure the floor isn't slippery and there's no furniture with sharp corners nearby. Stay close during the jumps and balancing until your little one gains confidence.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

Bottle Bowling

A homemade bowling alley with bottles to knock down by rolling a sock ball.

PREP

MATERIALS


 20-30
min


Materials

- Six empty plastic bottles
- A pair of socks (to make the ball)
- Water or a little sand to weigh down the bottles (optional)
- Tape to mark the throwing line (optional)

Prep ahead

Gather six matching bottles. If you want them to stand up better, add an inch of water or sand. Make a ball by stuffing one sock inside another and shaping it into a soft ball.

Step by step

- 1 Set up the bottles together in a triangle, like at a bowling alley.
- 2 Mark the throwing line with tape or a pillow.
- 3 Take turns rolling the sock ball to knock down the bottles.
- 4 Count together how many fell with each roll.
- 5 Set the pins back up and play several rounds, taking turns.

What it builds: Improves aim, hand-eye coordination, and throwing strength. Counting the fallen pins practices numbers, and waiting for a turn teaches patience.

Variations & tips: Paint or number the bottles and add up the points for the ones that fall. Move the line back little by little so the challenge grows with your little one's skill.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

The Pretend Store

Set up a little shop at home to buy and sell, with cardboard coins and lots of imagination.

PREP

MATERIALS



20-30
min



Materials

- Items from around the house (fruit, empty boxes, toys, books)
- Cardboard to cut out coins and bills
- Scissors (for the grown-up)
- Pencils to draw the prices
- A table or box as a counter, and a bag for the shopping

Prep ahead

The grown-up cuts out several round coins and cardboard bills. Choose a few safe items together and set up the counter. Draw simple prices (1, 2, 3) on little slips of paper.

Step by step

- 1 Arrange the items on the counter with their price next to each one.
- 2 Divide the cardboard coins between the shopper and the shopkeeper.
- 3 Take turns: one plays shopkeeper and says hello, the other picks what to buy.
- 4 Count the coins together to pay and make change.
- 5 Switch roles so your little one gets to be both seller and shopper.

What it builds: Encourages pretend play and language by acting out real-life situations. Counting coins introduces numbers, and taking turns teaches courtesy and waiting.

Variations & tips: Turn the store into a bakery, fruit stand, or toy shop. Add a notepad so the shopkeeper can 'write down' the orders with scribbles.

Safety: The grown-up handles the scissors. Use safe foods and avoid small pieces that could go in the mouth during play.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •



GAME

Ages 3–5

The Color Hunt

A race around the house to find objects of every color and fill up the rainbow basket.

PREP

MATERIALS



20-30
min



Materials

- Sheets of paper in several colors (or crayons to draw color blobs)
- A basket, box, or bag
- Household objects in different colors

Prep ahead

Prepare one sheet for each color, or draw big blobs of color on paper. Clear a spot where the found objects can be collected. Explain the rules simply.

Step by step

- 1 Show them a color card, for example red.
- 2 Ask them to search the house for something that color and bring it to the basket.
- 3 When they find it, name the object and its color together out loud.
- 4 Move on to the next color and repeat until you've gathered one object of each.
- 5 At the end, sort the objects by color and celebrate the rainbow you built.

What it builds: Reinforces color recognition and sorting. Searching around the house gets the body moving and works visual memory and attention, all while building vocabulary.

Variations & tips: Play against the clock to find everything red in one minute. Hunt by shape (round, square) or by size instead of color.

Safety: Agree on which areas and objects are allowed. Keep your little one away from anything fragile, hot, or with small parts during the hunt.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

The Faces of Feelings

Draw and act out happy, angry, and scared faces to learn to name what you're feeling.

PREP

MATERIALS



20-30
min



Materials

- Sheets of paper or paper plates
- Pencils or crayons
- A mirror (optional)

Prep ahead

Gather paper and crayons on the table. If you have a mirror handy, set it nearby so your little one can see their own face. Pick a calm moment, with no rush.

Step by step

- 1 Draw several faces together: a happy one, an angry one, a sad one, a scared one.
- 2 Name each emotion as you draw it: 'this face is happy.'
- 3 Take turns picking a face and making that same expression in the mirror.
- 4 Tell your little one about a time you felt that emotion, and ask about theirs.
- 5 Keep the faces where you can see them, so on other days you can point and say 'this is how I feel today.'

What it builds: Helps kids recognize and name their emotions, the first step to managing them. Talking calmly about feelings builds security and strengthens the bond between grown-up and child.

Variations & tips: Read stories together and pause to guess how each character feels. Make a 'face of the day' every morning to start the day chatting about moods.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

Potato and Veggie Stamps

Stamp shapes over and over with stamps made from potatoes, corks, and veggies.

PREP

MATERIALS



20-30
min



Materials

- A potato, a carrot, or a piece of celery
- Tempera paint or homemade paint (water, flour, and food coloring)
- A plate for the paint
- Sheets of paper
- Knife (grown-up only)

Prep ahead

The grown-up cuts the potato in half and, if they like, carves a simple shape into the surface (a heart, a star). Pour a little paint onto the plate and lay out the sheets of paper.

Step by step

- 1 Show them how to dip the stamp into the paint on the plate.
- 2 Have them press the stamp onto the paper and lift it carefully to see the shape.
- 3 Repeat with different stamps and colors to fill the page with prints.
- 4 Encourage them to mix shapes and colors freely, without aiming for a perfect picture.
- 5 Let the artwork dry and hang it where your little one can see it.

What it builds: Builds fine motor skills and coordination through pressing and lifting. Stamping invites experimenting with patterns, colors, and rhythm, valuing free creation.

Variations & tips: Use corks, bottle caps, or textured pieces as stamps. Stamp a repeating pattern to decorate a card or homemade wrapping paper.

Safety: Only the grown-up cuts or carves the veggies with the knife. Use non-toxic paint and supervise so your little one doesn't put it in their mouth; protect the table and clothes.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

Leaf and Flower Collage

Glue leaves, petals, and twigs to create a picture full of nature's treasures.

PREP

MATERIALS



20-30
min



Materials

- Leaves, flowers, twigs, and seeds collected outside
- A piece of poster board or cardboard
- A glue stick or white glue
- Pencils or crayons

Prep ahead

Take a short walk in the park, the yard, or down the street and gather leaves, petals, and twigs you both like. Back home, clear the table and keep the glue within reach.

Step by step

- 1 Empty the collected treasures onto the table and look at them together: colors, shapes, and smells.
- 2 Let your little one choose where each piece goes on the board.
- 3 Help them add glue and stick down the leaves and flowers one by one.
- 4 Have them finish the picture with crayon drawings all around.
- 5 Let the collage dry and make up a title for it together.

What it builds: Connects your little one with nature and its shapes. Gathering, choosing, and gluing works fine motor skills, observation, and creativity while composing a work of their own.

Variations & tips: Make one collage per season, with each one's typical colors. Line up the leaves from smallest to largest before gluing them down.

Safety: Collect only things from the ground, without pulling up plants, and avoid anything with thorns or prickles. Wash hands after the walk and when you're done.

 **We did it!**

Date: _____ ☆ ☆ ☆ ☆ ☆

What happened? What was the most fun?

Draw what happened:

Backyard Bug Safari

Head out as explorers with a cardboard magnifying glass to discover the little bugs living near home.

PREP

MATERIALS



20-30
min



Materials

- A clear cup or jar for observing (no closed lid)
- Cardboard to make a toy magnifying glass
- Scissors (for the grown-up)
- Paper and pencils to draw what you find

Prep ahead

The grown-up cuts a magnifying glass shape out of cardboard for pretend exploring. Pick a spot with dirt, plants, or flower pots, in the yard, the park, or on the balcony. Bring the cup and the paper.

Step by step

- 1 Walk slowly, checking the ground, leaves, and cracks with the cardboard magnifying glass.
- 2 When you spot a bug, watch it calmly without touching it: an ant, a ladybug, a snail.
- 3 If you want a closer look, cover it with the cup for a moment and then set it free.
- 4 Count how many legs it has, what color it is, and where it's headed.
- 5 Back home, draw the favorite bug from the outing and give it a name.

What it builds: Sparks curiosity about living things and respect for nature. Watching and describing bugs builds attention, language, and first ideas about animals.

Variations & tips: Make a list of the bugs you found and look for them on another outing. Observe plants, rocks, or clouds too, to widen the exploration.

Safety: The grown-up handles the scissors. Don't touch bugs that sting or plants with thorns; observe without harming them and always put them back where they were. Wash hands when you're done.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

EXPERIMENT

Ages 3–5

Oil and Water Won't Mix

Two liquids that refuse to blend and form magical bubbles inside the glass.

PREP

MATERIALS



20-30
min



Materials

- A tall clear glass
- Tap water
- Cooking oil (half a cup)
- Food coloring or a few drops of beet juice
- A spoon for stirring

Prep ahead

Fill the glass halfway with tap water and set it on the table. Keep the oil and food coloring within reach. Cover the surface with a tablecloth or newspaper in case anything spills.

Step by step

- 1 Ask your little one to look at the water, then add a few drops of food coloring together to tint it.
- 2 Now slowly pour the oil into the glass and watch what happens.
- 3 Ask them where the oil ended up: on top of the water or underneath?
- 4 Stir with the spoon and watch bubbles form and then separate again.
- 5 Let the glass sit still for a while and watch the oil rise back to the top.

What it builds: Introduces the idea that some liquids don't mix and that oil is lighter than water, so it floats. It builds observation, comparison, and vocabulary for describing what they see.

Variations & tips: Add half an effervescent tablet to see bubbles rise and fall like a lava lamp. Try it with warm and cold water to compare.

Safety: Oil makes the floor slippery if it spills: clean it up right away. It's not for drinking. Do the activity sitting down and on an easy-to-clean surface.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

 EXPERIMENT

Ages 3–5

White Flowers That Change Color

A white flower that wakes up painted in color, as if by magic.

PREP

MATERIALS



20-30
min



Materials

- One or two white flowers (carnation or daisy)
- Clear glasses
- Tap water
- Food coloring in several colors
- A spoon

Prep ahead

Fill each glass halfway with water and add plenty of food coloring to each one, one color per glass, so the water is nice and vivid. Have the flowers ready with their stems trimmed.

Step by step

- 1 Show your little one the white flowers and ask what color they think they'll turn.
- 2 Place one flower in each glass of colored water and leave them in a quiet spot.
- 3 Watch the flower for a while: nothing happens yet, you have to be patient.
- 4 The next day, look together at how the petals took on the color of the water.
- 5 Talk about which color changed the most and celebrate the discovery.

What it builds: Shows that plants drink water through their stems and carry it all the way to the petals. It teaches waiting, observing changes over time, and making simple predictions.

Variations & tips: Split a stem in two and put each half in a different color to get a two-toned flower. Use celery or a white cabbage leaf to watch the same thing happen.

Safety: Food coloring stains clothes and hands: wear an apron. It's not for drinking. The grown-up cuts the stems.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

 EXPERIMENT

Ages 3–5

The Candle Under the Glass

A lit candle that goes out all by itself when we trap its air under a glass.

PREP

MATERIALS


 20-30
min


Materials

- A small candle on a plate
- A glass (taller than the candle)
- Matches or a lighter (grown-up only)
- A shallow bowl with a little water (optional)

Prep ahead

Set the candle firmly on the plate, on a cleared table and away from anything that could catch fire. Keep the glass within reach. The grown-up is the only one who lights and handles the flame.

Step by step

- 1 The grown-up lights the candle while your little one watches the flame from a distance.
- 2 Ask what they think will happen if we cover the candle with the glass.
- 3 The grown-up covers the candle with the glass upside down, and you wait a few seconds together.
- 4 Watch the flame shrink and go out on its own without anyone blowing.
- 5 Talk about what you saw and repeat the experiment if they want to watch it again.

What it builds: Shows in a very simple way that fire needs air to keep burning. It builds observation, prediction, and respect for fire with a grown-up close by.

Variations & tips: Use glasses of different sizes and count how many seconds each takes to snuff out the flame. Set the candle on a plate with water and watch the water rise a little inside the glass.

Safety: This activity involves fire: only the grown-up lights and handles the candle, and your little one watches from a distance. Tie back long hair, move paper away, and keep water nearby. Never leave the candle unattended.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

 EXPERIMENT

Ages 3–5

The Balloon That Blows Itself Up

A balloon that inflates without any blowing, pushed by the bubbles of a magic mixture.

PREP

MATERIALS


 20-30
min


Materials

- A balloon
- A narrow-necked plastic bottle
- Baking soda (2 tablespoons)
- Vinegar (half a cup)
- A funnel or a paper cone

Prep ahead

Using the funnel, the grown-up pours the vinegar into the bottle. Separately, help your little one spoon the baking soda into the balloon using the same funnel, without letting them mix yet.

Step by step

- 1 Carefully stretch the mouth of the balloon (with the baking soda inside) over the neck of the bottle.
- 2 Ask your little one what they think will happen when the baking soda falls into the vinegar.
- 3 Count 'three, two, one' and lift the balloon so the baking soda drops in.
- 4 Watch lots of bubbles appear as the balloon starts inflating all by itself.
- 5 When it stops, talk about what you saw and, if you like, repeat it with another balloon.

What it builds: Shows that mixing baking soda and vinegar creates a gas that takes up space and inflates the balloon. It builds anticipation, observation, and the idea that air takes up room.

Variations & tips: Try more or less baking soda to see if the balloon grows bigger. Use bottles of different sizes and compare.

Safety: Vinegar stings if it gets in the eyes: supervise the mixing. It's not for drinking. Do the activity on an easy-to-clean surface.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

 EXPERIMENT

Ages 3–5

Seeds Sprouting in Cotton

Plant beans in cotton and discover, day by day, how they grow a root and a tiny leaf.

PREP

MATERIALS



20-30
min



Materials

- A clear jar or cup
- Cotton balls
- Dried beans or lentils
- Tap water
- A label or slip of paper to mark the date

Prep ahead

Put a nice layer of cotton inside the jar and moisten it with a little tap water, so it's damp but not flooded. Have the seeds ready on a small plate.

Step by step

- 1 Ask your little one to tuck a few seeds between the cotton and the glass, so you can see them.
- 2 Dampen the cotton with a little water and set the jar near a bright window.
- 3 Check the jar every day and add water if the cotton looks dry.
- 4 Watch how a white root appears first, and then a little stem with leaves.
- 5 Talk about how the plant grew and celebrate every new change.

What it builds: Shows that a seed needs water and light to sprout and become a plant. It teaches caring for something, observing patiently, and noticing changes day by day.

Variations & tips: Put one jar in the sun and another in a dark spot to compare. When the plant gets big, transplant it into a flower pot with soil.

Safety: Raw seeds are not for eating; make sure your little one doesn't put them in their mouth. Wash hands when you're done.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

EXPERIMENT

Ages 3–5

The Magnet and Its Hidden Treasures

An explorer magnet that discovers which things around the house stick to it and which don't.

PREP

MATERIALS



20-30
min



Materials

- A magnet (a strong fridge magnet works)
- A basket of assorted objects (metal spoon, ballpoint pen, wooden stick, key, plastic toy)
- Two boxes or plates for sorting

Prep ahead

Gather several safe objects in the basket: some metal and others wood, plastic, or fabric. Set out two plates: one for the things that stick and one for the things that don't.

Step by step

- 1 Show your little one the magnet and tell them it's going hunting for treasures that stick.
- 2 Take an object from the basket and bring the magnet close to see if it sticks.
- 3 Ask: did it stick or not? Then put the object on the matching plate.
- 4 Repeat with all the objects until they're sorted into two groups.
- 5 Look at the two plates together and talk about what the sticky ones had in common.

What it builds: Introduces the idea that magnets attract some metals and not other materials. It encourages sorting, predicting, and observing the qualities of objects.

Variations & tips: Take the magnet on a walk around the house looking for surfaces it sticks to, like the fridge. Hide metal objects in a bucket of rice and rescue them with the magnet.

Safety: Small magnets are dangerous if swallowed: use a big one and keep an eye on your little one. Keep the magnet away from electronics.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

 EXPERIMENT

Ages 3–5

A Rainbow from Water and a Mirror

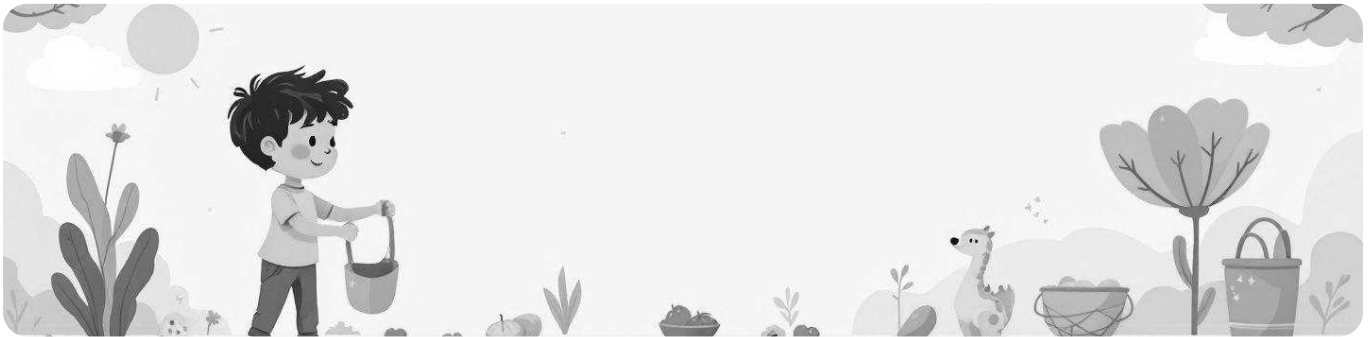
Catch sunlight with a mirror underwater and paint the wall with colors.

PREP

MATERIALS



20-30
min



Materials

- A deep dish or bucket of water
- A small hand mirror
- A sunny spot or a flashlight
- A wall or white poster board

Prep ahead

Fill the dish with tap water and place it where sunlight comes in, near a light-colored wall. If there's no sun, have a flashlight ready to use in a darker room.

Step by step

- 1 Place the mirror in the water, leaning at an angle against the edge of the dish.
- 2 Move the dish or the mirror until sunlight hits the mirror.
- 3 Look together for the rainbow of colors reflected on the wall.
- 4 Name the colors you see and move the mirror to make the rainbow dance.
- 5 Talk about how the water and light made the colors appear.

What it builds: Shows that white light hides many colors and that water separates them to form a rainbow. It encourages observation, naming colors, and curiosity about light.

Variations & tips: Use a flashlight in a dark room to make the rainbow without sun. Blow gently on the water to watch the rainbow ripple and sway.

Safety: Never look directly at the sun or point the reflected light at anyone's eyes. Dry the floor if it gets wet to avoid slips.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:



Ages 3–5

Nature Treasure Walk

Head out with a basket to gather leaves, pebbles, and sticks and build a collection all their own.

PREP

MATERIALS



20-30
min



Materials

- A basket or small bucket
- Bags or boxes to sort the treasures
- Comfy clothes for going out
- A toy magnifying glass (optional)

Prep ahead

Pick a spot with trees and plants: the backyard, a park, or a tree-lined sidewalk. Bring the basket and, before heading out, chat about what kinds of things you can collect.

Step by step

- 1 Walk without hurrying, looking at the ground, the trees, and the bushes.
- 2 Gather leaves of different sizes, pebbles, sticks, and fallen flowers in the basket.
- 3 Look at each treasure: ask your little one what color it is and whether it's smooth or rough.
- 4 Back home, sort the treasures by color, size, or shape.
- 5 Talk about the walk and keep the collection in a special spot.

What it builds: Brings your little one closer to nature and its textures, colors, and shapes. Collecting and sorting builds observation, classification, and care for the world around them.

Variations & tips: Glue the treasures onto poster board to make a nature mural. Repeat the walk in another season and compare what changed.

Safety: Don't collect plants with thorns or touch bugs that sting. Wash hands when you get back, and nothing goes in the mouth.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

 SENSORY

Ages 3–5

Fluffy Foam Painting

Paint with clouds of squishy foam that feel cool and fluffy in their hands.

PREP

MATERIALS



20-30
min



Materials

- Shaving cream or foam whipped up from a little dish soap
- Tempera paint in several colors
- A tray or flat dish
- Spoons and a paintbrush
- An apron or clothes that can get messy

Prep ahead

Spread a thick layer of foam on the tray. Have the tempera paint ready in little cups and an apron for your little one. Cover the table with a wipeable tablecloth or newspaper.

Step by step

- 1 Invite your little one to touch the foam and talk about how it feels: soft, cool, squishy.
- 2 Drip a few drops of colored tempera paint onto the foam.
- 3 Mix with fingers or a spoon and watch new colors appear.
- 4 Draw pictures with a finger in the colorful foam.
- 5 Celebrate every creation and let your little one experiment freely.

What it builds: Stimulates the sense of touch and free sensory exploration. Mixing colors and drawing with their fingers nurtures creativity and hand coordination.

Variations & tips: Press a sheet of paper onto the colored foam and lift it to leave a printed impression. Add a little white glue to the foam so the artwork hardens as it dries.

Safety: The foam isn't for eating or for the eyes: make sure your little one keeps their hands away from their face. Wash hands well when you're done.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

The At-Home Treasure Hunt

Follow clues all around the house until you find a hidden treasure at the end of the trail.

PREP

MATERIALS



20-30
min



Materials

- Simple objects to hide as clues (toys, drawings)
- A final treasure (a yummy piece of fruit, a sticker, a giant hug)
- A basket to hold what they find

Prep ahead

Before you start, hide several objects or drawings around the house as clues: each one points to where the next one is. Leave the treasure in the last hiding spot, for example in the living room.

Step by step

- 1 Tell your little one there's a hidden treasure and hand them the first clue.
- 2 Give clues in simple words: 'look near the fridge' or 'check under the pillow.'
- 3 Go along as they search for each object and celebrate every find together.
- 4 Follow the clues one after another all around the house.
- 5 When they find the final treasure, cheer the victory with a round of applause.

What it builds: Builds attention, memory, and the ability to follow directions. Searching and following clues encourages language, a sense of space, and perseverance.

Variations & tips: Make picture clues so your little one can read them on their own. Hide the treasure in the yard for an outdoor version.

Safety: Hide the clues in safe spots, with no climbing on chairs or reaching into high drawers. Stay with your little one for the whole hunt.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

The Calm-Down Jar

A jar of glitter that drifts down slowly and helps you breathe and settle when you're upset.

PREP

MATERIALS



20-30
min



Materials

- A plastic jar with a tight-fitting lid
- Warm tap water
- White glue or glycerin
- Glitter, sequins, or sparkles
- Food coloring (optional)

Prep ahead

Have the jar clean and dry. Get the warm water, white glue, and glitter within reach. The grown-up closes and seals the lid tightly at the end so it can't come open.

Step by step

- 1 Help your little one pour warm water into the jar until it's almost full.
- 2 Add a squeeze of white glue, the glitter, and a drop of food coloring.
- 3 The grown-up closes the lid very tightly and seals it so it holds firm.
- 4 Shake the jar together and watch the glitter swirl up and drift slowly down.
- 5 Breathe slowly while watching the glitter settle, and talk about how you're feeling.

What it builds: Gives kids a tool for recognizing and calming big emotions. Watching the glitter and breathing slowly helps your little one settle down when they're angry or frustrated.

Variations & tips: Make a jar for each emotion in different colors. Use the jar as a waiting timer: when the glitter settles, it's time to move on.

Safety: The grown-up seals the lid so the jar can't open. Use a plastic jar, not glass, so it won't break.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

Bubble-Print Paintings

Blow colorful bubbles that pop on the paper and leave amazing pictures behind.

PREP

MATERIALS



20-30
min



Materials

- One cup per color
- Dish soap and a little water
- Tempera paint in several colors
- Straws (one per color)
- Sheets of thick paper

Prep ahead

In each cup, mix a little water, a squirt of dish soap, and plenty of tempera paint in one color. Put a straw in each cup. Cover the table and put an apron on your little one.

Step by step

- 1 Show them how to blow gently through the straw into the cup until lots of bubbles rise up.
- 2 Remind them to blow, never suck, so they don't swallow the mixture.
- 3 When the bubbles overflow the cup, press the paper on top so they pop onto the page.
- 4 Repeat with other colors on different parts of the sheet.
- 5 Let the painting dry and celebrate the round little prints left behind.

What it builds: Encourages creativity and breath control through blowing. Watching the bubbles leave shapes on the paper sparks wonder and a love of making art.

Variations & tips: Mix two colors on the same sheet to discover new colors. Cut out the round shapes and put together a collage.

Safety: Your little one must blow, never sip, the dish soap mixture. Stay with them the whole time and wash hands when you're done.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

 EXPERIMENT

Ages 3–5

Magic Rainbow Milk

The colors shoot off and dance across the milk all on their own the moment you touch it with soap.

PREP

MATERIALS



20-30
min



Materials

- A shallow bowl or white dish
- Whole milk (whatever's in the fridge)
- Food coloring in several colors
- Dish soap in a separate little bowl
- A straw or a cotton swab

Prep ahead

Take the milk out of the fridge and pour a layer that covers the bottom of the dish. Let it settle without moving it. Add a few drops of each food coloring to different spots on the milk. Keep the dish soap in a little bowl next to you and cover the table in case of spills.

Step by step

- 1 Ask your little one to look at the little dots of color sitting still on the milk.
- 2 Together, dip the tip of the straw or cotton swab into the dish soap.
- 3 Touch the tip to the center of the milk, without stirring, and watch what happens.
- 4 Watch the colors race toward the edges and swirl together in little whirlpools.
- 5 Try touching other spots on the milk to create new bursts of color.
- 6 Talk about which color they liked best and where it ran off to.

What it builds: Introduces the idea that soap pushes the fat in the milk, which is why the colors move on their own. It builds observation, anticipating what will happen next, and movement vocabulary like push, escape, and spin.

Variations & tips: Try skim milk to compare whether the colors move the same or less. Press a white sheet of paper on top at the end to save a print of the swirled colors.

Safety: Milk with dish soap is not for drinking: it's just for watching. Stay with your little one the whole time and wash hands when you're done. Wipe up spills on the floor right away so it doesn't get slippery.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

 EXPERIMENT

Ages 3–5

The Pepper That Runs Away

Pepper floats peacefully on the water, then scatters in a flash the second a soapy finger comes near.

PREP

MATERIALS


 20-30
min


Materials

- A shallow bowl or dish
- Tap water
- Ground pepper
- A little dish soap in a small bowl

Prep ahead

Fill the dish with about an inch of tap water and leave it still on the table so the water settles. Keep the pepper and the small bowl of dish soap within reach. Cover the table in case of splashes.

Step by step

- 1 Ask your little one to sprinkle pepper across the whole surface of the water.
- 2 Watch how the pepper floats quietly on top of the water.
- 3 Together, dip a finger in plain water and touch it: almost nothing happens.
- 4 Now dip a fingertip in the dish soap and touch the center of the water again.
- 5 Watch the pepper shoot off toward the edges in an instant.
- 6 Talk about why the pepper ran away so fast when the soap showed up.

What it builds: Introduces the idea that water has an invisible skin on its surface and that soap breaks it, making the pepper flee. It builds observation, comparing the finger with and without soap, and wonder at a sudden change.

Variations & tips: Try salt or oregano instead of pepper to see if they move the same way. Repeat with clean water to discover the trick only works when the water has no soap in it yet.

Safety: Soapy water is not for drinking or for the eyes: wash hands when you're done. Stay with your little one for the whole activity and dry the floor if water spills to avoid slips.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •



Ages 6-8

Create, build and discover

Ages 6–8

Create, build and discover

They can read, count and follow multi-step instructions now. They love challenges, projects with a result they can show off, and games with rules. It's the perfect age for bigger experiments, crafts and homemade board games. Let them make decisions and solve problems: getting it wrong is learning too. Your role is guide and teammate.

26

ACTIVITIES

12

SCIENCE EXPERIMENTS

13

READY TO START NOW

In this chapter

● The Magic Tower of Liquids	30-45 min	● Homemade Lava Lamp	30-45 min
● Balloon Rocket on a String	30-45 min	● Secret Messages in Invisible Ink	30-45 min
● Rainbow Salt Mandala	30-45 min	● The Goo That's Liquid and Solid at Once	30-45 min
● The Mystery Touch Box	30-45 min	● The Hidden Colors Inside Markers	30-45 min
● Homemade Obstacle Course	30-45 min	● The Backyard Sundial	30-45 min
● Homemade Bottle Bowling	30-45 min	● The Paper Bridge That Holds Weight	30-45 min
● Invent Your Own Board Game	30-45 min	● The Plant That Chases the Light	30-45 min
● Treasure Hunt with Clues	30-45 min	● The Freeze-Statue Race	30-45 min
● The Calm-Down Jar	30-45 min	● Veggie Stamp Art	30-45 min
● Cardboard Puppets and a Puppet Show	30-45 min	● The Feelings Thermometer	30-45 min
● Nature Explorer's Journal	30-45 min	● Guess the Texture with Your Eyes Closed	30-45 min
● The Magic Electric Balloon	30-45 min	● The Fireproof Balloon	30-45 min
● Salt Crystals That Grow on Their Own	30-45 min	● The Toy Parachute	30-45 min

 EXPERIMENT

Ages 6–8

The Magic Tower of Liquids

Stack honey, water, and oil in a single glass and discover why they never mix.

PREP

MATERIALS


 30-45
min


Materials

- A tall, clear glass or jar
- Honey or syrup
- Water
- Cooking oil
- Food coloring or watered-down tempera paint (optional)
- A spoon
- Small objects to test: a cork, a grape, a metal nut, a little piece of cardboard

Prep ahead

Gather the three liquids on the table. If you want more color, mix a few drops of food coloring into the water. Set the empty glass in the center and let your child measure out equal amounts of each liquid into three small cups.

Step by step

- 1 Ask your child which liquid they think is the 'heaviest.' Write down the prediction before you start.
- 2 Pour the honey into the bottom of the glass first, slowly and right in the center.
- 3 Add the water by letting it slide down the side of the glass or off the spoon so it lands gently.
- 4 Pour in the oil last the same way, and watch three separate layers form.
- 5 Carefully drop in the small objects and see which layer each one ends up floating in.
- 6 Compare the results with your prediction from the beginning and talk about the surprises.

What it builds: Introduces the idea of density: denser liquids and objects sink below less dense ones. Your child practices observing, predicting, and testing like a real scientist.

Variations & tips: Try adding tinted rubbing alcohol as a fourth floating layer, or dissolve salt into one batch of water to create two water layers of different colors.

Safety: Honey and oil stain: cover the table. Don't taste any of the liquids in the experiment, and wash your hands when you're done.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

 EXPERIMENT

Ages 6–8

Balloon Rocket on a String

Build an invisible track and send a balloon zooming across the room like a rocket.

PREP

MATERIALS



30-45
min



Materials

- A long balloon
- A long piece of string or yarn (10 to 15 feet)
- A straw or a rolled-up paper tube
- Tape
- A clothespin (optional)

Prep ahead

Find two spots across the room to tie the string to, such as the back of a chair and a door handle. Thread the string through the straw before tying the second end, and pull it nice and tight.

Step by step

- 1 Blow up the balloon without tying it, and pinch the neck closed with your fingers or a clothespin.
- 2 With an adult's help, tape the balloon to the straw with two pieces of tape, pointing the neck toward one end of the string.
- 3 Slide the balloon to the start of the track so it has the whole run ahead of it.
- 4 Count down: three, two, one, and let go of the balloon's neck.
- 5 Watch how the escaping air pushes the balloon to the other end of the string.
- 6 Repeat with different amounts of air or a tilted string, and compare which rocket goes the farthest.

What it builds: Shows the principle of action and reaction: the air rushes out in one direction and pushes the balloon in the opposite one, just like real rockets. Your child experiments with variables and measures results.

Variations & tips: Set up a race with two parallel strings and two balloons, or hang a small paper weight from the rocket to see how much cargo it can carry.

Safety: An adult should inflate the balloon and tie the string at safe heights. Keep broken balloons away from small children, since they're dangerous if put in the mouth.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

Rainbow Salt Mandala

Color salt with chalk and create a colorful mosaic that fits in the palm of your hand.

PREP

MATERIALS



30-45
min



Materials

- Table salt
- Colored chalk
- Several small cups or plates
- A sheet of cardboard or poster board
- Liquid glue
- A pencil

Prep ahead

Divide a little salt into each cup. Cover the table with newspaper, because colored salt scatters easily. Together with your child, draw a simple mandala in pencil on the cardboard: circles, petals, and stripes.

Step by step

- 1 Rub a piece of colored chalk against the salt in one cup until the salt is completely tinted.
- 2 Repeat with other colors in the remaining cups so you have a little palette of salt.
- 3 Spread glue with your finger or a brush over one section of the drawing at a time.
- 4 Sprinkle the colored salt over the glue and shake the extra off onto the newspaper.
- 5 Keep going section by section, switching colors, until the whole mandala is complete.
- 6 Let the mandala dry flat for a while, then hold it up to the light.

What it builds: Builds fine motor skills, color mixing, and patience. Designing a symmetrical pattern helps kids plan ahead and stay focused on a project with a finished result.

Variations & tips: Use sand instead of salt, or layer the colored salt in a clear jar to make a decorative keepsake.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

The Mystery Touch Box

Reach in without looking and guess which hidden object you're touching.

PREP

MATERIALS



30-45
min



Materials

- A cardboard box with a lid
- An old sock or a piece of fabric
- Scissors
- Household objects with different textures and shapes (a sponge, a pinecone, a spoon, a ball, a key, a feather)
- Paper and pencil to keep score

Prep ahead

An adult cuts a hand-sized hole in one side of the box and tapes the cut-off sock around it, so a hand can reach in but no one can peek inside. Secretly gather several safe objects.

Step by step

- 1 While one player closes their eyes, another places an object inside the box.
- 2 The player reaches through the hole and feels the object without pulling it out or looking.
- 3 They describe out loud what they feel: whether it's hard or soft, cold, rough, or smooth.
- 4 After describing it, they say their guess and check whether they got it right.
- 5 Score a point for each correct guess, and take turns hiding and guessing.
- 6 At the end, talk about which touch clues were the most helpful.

What it builds: Sharpens the sense of touch and descriptive vocabulary. Your child learns to reason from clues and to put sensations into words without relying on sight.

Variations & tips: Try it with only fruits and vegetables, or add a timer to guess against the clock. You can also use your ears by shaking the box before touching.

Safety: An adult should do the cutting with scissors. Avoid sharp or very small objects inside the box.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

Homemade Obstacle Course

Turn the living room or backyard into a course of jumps, tunnels, and balancing acts.

PREP

MATERIALS



30-45
min



Materials

- Cushions and pillows
- Chairs
- A blanket or sheet
- String or yarn
- Plastic cups or empty bottles
- A stopwatch or clock

Prep ahead

Clear a safe area and move anything breakable out of the way. Decide with your child how many stations the course will have, and gather the materials in one spot so you can build them together.

Step by step

- 1 Design the route together by drawing it on paper: where it starts and where it ends.
- 2 Build a tunnel by draping a blanket over two chairs to crawl under.
- 3 Place cushions on the floor as islands to jump from one to the next without touching the ground.
- 4 Stretch a string between two pieces of furniture at a low height to duck under without touching it.
- 5 Line up the cups to weave around in a zigzag as you run between them.
- 6 Time each lap and try to beat your time on every attempt.

What it builds: Develops coordination, balance, and movement planning. Designing the course and beating their own time builds effort and self-confidence.

Variations & tips: Add an aiming challenge by tossing a sock ball into a bucket at the end, or run the course in pairs holding hands.

Safety: Keep hard corners and slippery floors out of the course. Keep the string low to avoid dangerous trips, and supervise the jumps.

 **We did it!**

Date: _____ ☆ ☆ ☆ ☆ ☆

What happened? What was the most fun?

Draw what happened:

 ACTIVE PLAY

Ages 6–8

Homemade Bottle Bowling

Build your own bowling alley out of bottles and roll to knock them all down.

PREP

MATERIALS



30-45
min



Materials

- Six to ten matching empty plastic bottles
- A little water, sand, or rice
- A ball or a ball of rolled-up socks
- Tempera paint or colored paper for decorating
- Tape to mark the floor
- Paper and pencil for keeping score

Prep ahead

Wash and dry the bottles. Pour an inch of water, sand, or rice into each one so they stand steady but still fall when hit. Decorate them with paint or by gluing on the numbers 1 through 10.

Step by step

- 1 Set the bottles up in a triangle, just like in a real bowling alley.
- 2 Mark a throwing line with tape a few steps away.
- 3 Each player rolls the ball to knock down as many bottles as they can.
- 4 Add up points based on the bottles knocked down or the numbers written on them.
- 5 Set the bottles back up and take a second roll each turn.
- 6 Keep score by rounds and crown a champion at the end.

What it builds: Improves aim, mental math from adding points, and turn-taking. Building and decorating the pins adds creativity before the game even begins.

Variations & tips: Move the line farther back to raise the difficulty, play blindfolded guided by someone's voice, or put different weights in the bottles to see which ones are hardest to knock over.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •



Ages 6–8

Invent Your Own Board Game

Draw a game board with your own rules and race to be the first one to the finish.

PREP

MATERIALS



30-45
min



Materials

- A large piece of poster board or cardboard
- Colored pencils or markers
- A die (or a handmade paper one)
- Game pieces: buttons, bottle caps, or different coins
- Slips of paper for writing challenges and prizes

Prep ahead

Sit down together with the poster board on the table. Agree on the game's theme before you start: a jungle, outer space, a castle. That will spark more ideas for inventing the squares.

Step by step

- 1 Draw a zigzag path of squares from a starting point to a finish line.
- 2 Number the squares and color a few special ones: move ahead, move back, or lose a turn.
- 3 Write fun challenges on slips of paper for certain squares, like jumping three times.
- 4 Each player picks a game piece and places it at the start.
- 5 Take turns rolling the die, moving the number of squares shown, and doing whatever each square says.
- 6 Whoever reaches the finish first while following all the invented rules wins.

What it builds: Sparks creativity, number reading, and cause-and-effect thinking through designing fair rules. Taking turns teaches kids to wait, win, and lose with good sportsmanship.

Variations & tips: Create surprise cards to draw when landing on certain squares, or make two paths that cross to add traps and shortcuts.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

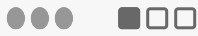
• • •

Treasure Hunt with Clues

Follow a trail of riddles through the house until you find the hidden treasure.

PREP

MATERIALS


 30-45
min


Materials

- Paper and pencil for writing the clues
- An envelope or box for the treasure
- A small prize: a treat, a drawing, or a craft
- Tape for hiding the clues

Prep ahead

While your child isn't looking, an adult writes five to eight clues. Each clue is a riddle that leads to the next hiding spot: 'Where cold things are kept' points to the fridge. Hide the treasure at the end of the trail and place the clues in reverse order.

Step by step

- 1 Hand your child the first clue and read the riddle together if needed.
- 2 Let them think about which spot in the house it means before running off to search.
- 3 At each hiding spot they'll find the next clue taped in place.
- 4 Encourage them to read out loud and reason through each riddle on their own.
- 5 Clue by clue, your child moves along until the treasure's final hiding spot.
- 6 When they find the treasure, celebrate and talk about which clue was the hardest.

What it builds: Strengthens reading comprehension, logic, and spatial awareness. Solving a chain of riddles builds perseverance and the satisfaction of reaching a goal step by step.

Variations & tips: Make picture clues instead of written ones for early readers, or hide the treasure outside, in the yard or at the park, for an outdoor version.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

The Calm-Down Jar

Create a glitter galaxy in a jar to help you settle down when you're upset.

PREP

MATERIALS



30-45
min



Materials

- A clear jar or bottle with a tight-fitting lid
- Warm water
- Clear liquid glue or glitter glue
- Fine glitter
- Food coloring (optional)
- Strong tape to seal the lid

Prep ahead

Gather the materials on a covered table. Talk with your child about big feelings first: explaining that we all get angry or nervous sometimes sets up the whole point of this activity.

Step by step

- 1 Fill the jar three-quarters full with warm water.
- 2 Add several spoonfuls of glue; the more glue, the more slowly the glitter will fall.
- 3 Pour in the glitter and, if you like, a drop of food coloring, then close the jar tightly.
- 4 An adult seals the lid with tape so it can't open or spill.
- 5 Shake the jar and quietly watch the glitter swirl and slowly settle.
- 6 Talk about how, just like the glitter, angry feelings settle down if we wait and breathe.

What it builds: Helps kids recognize and manage intense emotions. Watching the glitter settle goes hand in hand with slow breathing and teaches that emotions, like the storm in the jar, pass with time.

Variations & tips: Make several jars with colors tied to different emotions, or use the jar as a gentle timer for a one-minute calm-down break.

Safety: An adult should seal the lid carefully. Use a plastic jar if there's a risk it could fall and break.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

Cardboard Puppets and a Puppet Show

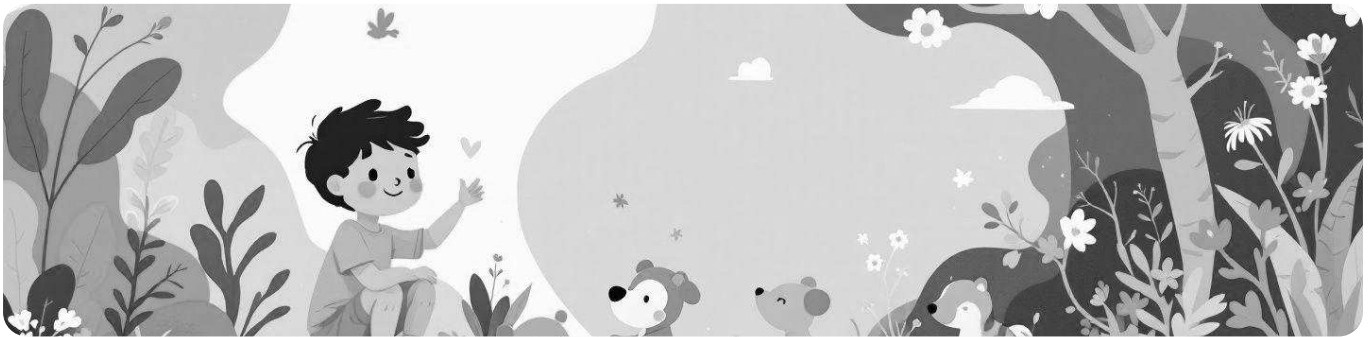
Bring cardboard characters to life and put on your very own puppet show.

PREP

MATERIALS



30-45
min



Materials

- Cardboard or poster board
- Safety scissors
- Craft sticks, pencils, or straws for the handles
- Tempera paint or markers
- Tape or glue
- A large box or a table for the stage

Prep ahead

Decide together on a simple story and which characters you'll need: maybe a hero, an animal, and a villain. Cover the table for painting and cut the cardboard into manageable pieces.

Step by step

- 1 Draw each character on the cardboard and cut out the shapes, with help if needed.
- 2 Paint and decorate the characters with faces, clothes, and fun details.
- 3 Tape a stick to the back of each figure so you can hold it up and move it around.
- 4 Turn a tipped-over box or the edge of a table draped with fabric into the stage.
- 5 Rehearse the story, giving each puppet its own voice.
- 6 Perform the show for the rest of the family like a real theater.

What it builds: Encourages creativity, language, and spoken expression. Making up and acting out a story develops imagination, narrative sequencing, and confidence speaking in front of others.

Variations & tips: Make shadow puppets by cutting out silhouettes and shining a flashlight behind a sheet, or record the show's audio to listen to later.

Safety: Have your child use safety scissors, and let an adult supervise the trickier cuts.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

Nature Explorer's Journal

Head outside with a magnifying glass and a notebook and become a nature explorer.

PREP

MATERIALS



30-45
min



Materials

- A notebook or several sheets of paper stapled together
- A pencil and colored pencils
- A magnifying glass (optional)
- A small bag for collecting treasures
- Tape for attaching finds

Prep ahead

Get the notebook ready by writing 'Explorer's Journal' and the date on the cover. Pick a nearby spot with plants: a park, the yard, or a sidewalk lined with trees. Dress comfortably for being outdoors.

Step by step

- 1 Walk slowly, taking your time to observe leaves, insects, rocks, and flowers.
- 2 Use the magnifying glass to look up close at the veins of a leaf or the legs of a bug.
- 3 Draw three things in the notebook that catch your eye during the walk.
- 4 Collect leaves or flowers that have fallen to the ground, without picking anything from living plants.
- 5 Tape the finds into the notebook and write next to them where and when you found them.
- 6 Back home, count how many different species you recorded.

What it builds: Awakens scientific curiosity and careful observation. By recording and sorting what they see, your child learns to notice details, compare, and respect nature.

Variations & tips: Repeat the walk in different seasons to compare the changes, or devote each day's page to a single thing, like only leaves or only rocks.

Safety: Don't touch thorny plants, mushrooms, or unfamiliar animals, and wash your hands when you get back home.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

 EXPERIMENT

Ages 6–8

The Magic Electric Balloon

Rub a balloon on your hair and make things move without touching them.

PREP

MATERIALS



30-45
min



Materials

- An inflated balloon
- Tiny scraps of paper
- A thin thread or a strand of hair
- A trickle of water from the tap
- An empty soda can

Prep ahead

Blow up the balloon and tie the knot. Cut up some very small paper scraps and scatter them on the table. This experiment works better on a dry day than a very humid one.

Step by step

- 1 Rub the balloon firmly against your hair or a wool sweater for a few seconds.
- 2 Bring the balloon close to the paper scraps without touching them, and watch them jump up toward it.
- 3 Charge the balloon again and hold it near a dangling thread to see it move on its own.
- 4 Turn the tap on just barely, so a thin stream of water forms, and bring the charged balloon close to bend the stream.
- 5 Lay the can on its side on the floor and slowly bring the balloon near to make it roll without touching it.
- 6 Talk about which objects reacted the most and why the balloon loses its charge when it touches other things.

What it builds: Introduces static electricity: rubbing the balloon charges it so it attracts light objects. Your child discovers an invisible force and learns to test it on different materials.

Variations & tips: Try charging two balloons and bringing them together to see how they push each other away, or try sticking the charged balloon to the wall.

Safety: This static electricity is harmless, but remember not to bring the balloon near anyone's eyes, and keep the water away from electrical outlets.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

 EXPERIMENT

Ages 6–8

Salt Crystals That Grow on Their Own

Mix up super-salty water and watch shiny crystals appear out of nowhere in just a few days.

PREP

MATERIALS



30-45

min



Materials

- Table salt
- A clear glass jar
- Hot water (prepared by an adult)
- A spoon
- A piece of yarn or thick string
- A pencil to rest across the jar
- A notebook for tracking the changes

Prep ahead

An adult heats the water and pours it into the jar. Clear a quiet spot where the jar can sit undisturbed for several days without anyone moving it. Have the notebook ready to record what happens each day.

Step by step

- 1 Add salt to the hot water one spoonful at a time, stirring, until no more will dissolve.
- 2 Tie the piece of yarn to the middle of the pencil and rest it across the mouth of the jar so the yarn hangs down into the water.
- 3 Leave the jar still in a spot where no one will touch or move it.
- 4 Each day, check the yarn and the bottom of the jar, and write or draw in the notebook what you see growing.
- 5 After several days, carefully lift out the crystal-covered yarn and take a close look.

What it builds: Shows how water can hold a lot of dissolved salt and, as it slowly evaporates, the salt comes back out and forms crystals. Your child practices patience and keeps records of an experiment that changes over time.

Variations & tips: Repeat the experiment with sugar in another jar and compare the shape of the crystals, or add a few drops of food coloring to grow colored crystals.

Safety: Only an adult handles the hot water. Don't taste the crystals or the salt water.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

. . .

 EXPERIMENT

Ages 6–8

Homemade Lava Lamp

Mix oil, water, and a fizzy tablet to watch colorful bubbles rise and fall like lava.

PREP

MATERIALS



30-45
min



Materials

- A clear bottle or jar
- Cooking oil
- Water
- Food coloring
- An effervescent tablet (the kind that fizzes)
- A notebook for writing things down

Prep ahead

Clear a table and cover it in case any oil spills. Fill the bottle three-quarters full with oil, and set aside a little water with a few drops of food coloring in a separate cup.

Step by step

- 1 Ask your child what they think will happen when the water meets the oil, and write down the prediction.
- 2 Pour the colored water into the bottle of oil and watch it separate into two layers.
- 3 Break the fizzy tablet into pieces and drop one into the bottle.
- 4 Watch colorful bubbles rise and fall, turning the bottle into a lava lamp.
- 5 When it calms down, add another piece to repeat the effect, and compare with your prediction.

What it builds: Teaches that oil and water don't mix because they have different densities, and that the tablet releases a gas that pushes the water upward in bubbles. Your child observes, predicts, and checks.

Variations & tips: Try it with warm water and then cold water to see if the bubbles change speed, or use two different food colorings in separate bottles.

Safety: Don't drink what's in the bottle. Oil is slippery and it stains: clean up spills right away.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

. . .

 EXPERIMENT

Ages 6–8

Secret Messages in Invisible Ink

Write a message in lemon juice and watch it appear like magic with a little heat.

PREP

MATERIALS



30-45
min



Materials

- Half a lemon, squeezed into a cup
- Sheets of white paper
- A fine paintbrush, a cotton swab, or a toothpick
- A safe heat source: a lit light bulb or an iron (used by an adult)
- A notebook for writing things down

Prep ahead

Squeeze the lemon into a cup and set the juice aside. Clear a well-lit table and lay out several sheets for practice runs. Decide together on a short, secret message to write.

Step by step

- 1 Dip the brush or cotton swab in the lemon juice and write the message on the paper.
- 2 Let the paper dry completely, until the message is invisible.
- 3 Ask someone else to try to read the blank-looking page; they'll see there's nothing there.
- 4 An adult holds the paper near the heat source, without burning it, until the letters turn brown and appear.
- 5 Read the secret message and note in the notebook how long it took to show up.

What it builds: Shows that lemon juice darkens with heat before the paper does, which is why the letters become visible. Your child discovers a reaction triggered by heat and gets a first taste of cause and effect.

Variations & tips: Try other juices like orange juice, or milk, and compare which one reveals the message best when heated.

Safety: Only an adult handles the heat source, to prevent burns and keep the paper from catching fire.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

 EXPERIMENT

Ages 6–8

The Goo That's Liquid and Solid at Once

Mix up cornstarch and water into a goo that hardens when you hit it and melts when you let it rest.

PREP

MATERIALS

● ○ ○

■ □ □


 30-45
min


Materials

- Cornstarch
- Water
- A large bowl
- A spoon
- Food coloring (optional)
- A notebook for writing things down

Prep ahead

Cover the table with paper or an old tablecloth, because this activity gets messy. Have the cornstarch and a cup of water within reach. If you want color, mix a few drops of food coloring into the water.

Step by step

- 1 Pour two cups of cornstarch into the bowl and add the water a little at a time while stirring.
- 2 Keep adding water until you get a mixture that looks liquid when you stir it gently.
- 3 Slap the surface quickly with your hand and notice how it feels hard, like a solid.
- 4 Grab some, squeeze it into a ball, then open your hand and watch it melt away.
- 5 Write in the notebook what happens when you touch it fast and what happens when you leave it still.

What it builds: Introduces a curious material that behaves like a solid when force is applied and like a liquid when it's at rest. Your child explores with their hands and describes what they observe in their own words.

Variations & tips: Change the amount of water to see how the texture varies, or split the goo into portions of different colors and mix them together.

Safety: Don't eat it. Throw the mixture in the trash, never down the drain, because it can clog the pipes.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

. . .

EXPERIMENT

Ages 6–8

The Hidden Colors Inside Markers

Discover that a black marker hides several secret colors that water can pull apart.

PREP

MATERIALS



30-45

min



Materials

- Markers in several colors (make sure they're washable)
- Coffee filters or paper napkins
- A glass with a little water
- Scissors
- A notebook for writing things down

Prep ahead

Cut the coffee filters into strips about a finger wide. Put a little water in the glass, just enough to cover the bottom. Clear the table and lay out the markers so you can choose which ones to test.

Step by step

- 1 Draw a thick marker dot on each strip, about two finger-widths from one end.
- 2 Ask your child which colors they think each marker is hiding, and write down the predictions.
- 3 Hang the strip inside the glass so only the bottom end touches the water, without wetting the colored dot.
- 4 Wait and watch as the water climbs up the paper and carries the colors along, separating them.
- 5 Compare the colors that appeared with what you predicted, and tape the strips into the notebook.

What it builds: Introduces chromatography: many colors are actually mixtures of others, and water separates them because each one travels up the paper at a different speed. Your child predicts, observes, and compares.

Variations & tips: Test markers of the same brand but different colors, or compare a brand-new marker with an almost worn-out one to see if the colors change.

Safety: Use washable markers. Don't drink the water in the glasses, and wash your hands when you're done.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

 EXPERIMENT

Ages 6–8

The Backyard Sundial

Build a clock out of a stick and a shadow, and learn to tell time by looking at the sky.

PREP

MATERIALS



30-45
min



Materials

- A straight stick or a long pencil
- A piece of cardboard or a paper plate
- Modeling clay or a little dirt to hold the stick steady
- A marker
- A clock to compare the time

Prep ahead

Pick a sunny day and a spot in the yard or on the sidewalk that gets sun all morning and afternoon. This activity means checking the clock every hour, so it's best to start early.

Step by step

- 1 Stand the stick up in the center of the cardboard with modeling clay so it stays firmly upright.
- 2 Set the cardboard in the sun and, at the top of every hour, mark with the marker where the tip of the shadow falls.
- 3 Next to each mark, write the time shown on the real clock.
- 4 Throughout the day, watch how the shadow moves and changes length.
- 5 The next day, try guessing the time by looking only at the shadow, then check against the clock.

What it builds: Shows that the sun appears to move across the sky and that shadows change position during the day, just like ancient clocks used. Your child connects watching the sky with measuring time.

Variations & tips: Compare the marks from a winter day with a summer day, or make a bigger sundial with chalk right on the patio or driveway.

Safety: Never look directly at the sun. Wear a hat and sunscreen if you'll be outside for a long time.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

 EXPERIMENT

Ages 6–8

The Paper Bridge That Holds Weight

Discover how a simple folded sheet of paper can hold up coins without collapsing.

PREP

MATERIALS

● ○ ○

■ □ □



30-45

min



Materials

- Several identical sheets of paper
- Two books or boxes of the same height
- Coins or dried beans to use as weights
- A notebook for writing things down

Prep ahead

Set the two books on the table with a gap between them, like the two banks of a river. Have the coins or beans in a pile, ready to use as test weights.

Step by step

- 1 Lay a flat sheet across the two books to form a bridge, and place coins on it until it sags.
- 2 Write down how many coins the flat bridge held before it fell.
- 3 Fold another sheet into an accordion, with lots of pleats, and use it as the new bridge.
- 4 Place coins on the folded bridge and count how many it holds now.
- 5 Compare the two results in the notebook and talk about why one held up more.

What it builds: Introduces an engineering idea: shape changes a material's strength, and folds spread the weight better than a flat sheet. Your child builds, tests, and measures to draw conclusions.

Variations & tips: Try other shapes, like rolling the paper into tubes, or widen the gap between the books to see how far the bridge can hold out.

Safety: Use coins or beans as weights, not heavy objects that could fall on someone's feet.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

The Plant That Chases the Light

Grow a bean and watch its stem lean toward the window all on its own, searching for the sun.

PREP

MATERIALS



30-45
min



Materials

- A dried bean or a fast-growing seed
- A clear jar or glass
- Cotton balls or paper napkins
- Water
- A spot near a window
- A notebook for writing things down

Prep ahead

Dampen the cotton with water, without letting it drip, and place it inside the jar. Pick a spot next to a window that gets good light during the day. Have the notebook ready to keep a record.

Step by step

- 1 Tuck the bean between the cotton and the glass of the jar so you can watch it grow.
- 2 Set the jar near the window and keep the cotton moist at all times.
- 3 Each day, observe and draw in the notebook how the root and stem appear.
- 4 Once the stem has grown, turn the jar so the plant faces away from the light.
- 5 Keep watching over the next few days as the stem leans back toward the window.

What it builds: Shows phototropism: plants grow toward the light they need to live. Your child cares for a living thing, records its changes, and discovers that plants react to their surroundings.

Variations & tips: Put another bean inside a dark box with a single hole for light and compare which way it grows, or plant the bean in a pot with soil once it has roots.

Safety: Wash your hands after handling soil and damp cotton. Don't put the seeds in your mouth.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

The Freeze-Statue Race

Sneak forward without getting caught moving: when the music stops, everyone freezes like statues.

PREP

MATERIALS



30-45
min



Materials

- A wide-open space, indoors or outside
- Something to mark the finish line: a rope, chalk, or a cushion
- Music, or someone to sing or clap a beat

Prep ahead

Clear the area of anything players could trip over. Mark a starting line at one end and the finish line at the other. Explain the rules to everyone before you begin.

Step by step

- 1 One person stands at the finish line with their back turned, playing the guard while the music plays.
- 2 While the music is on, everyone races toward the finish line as fast as they can.
- 3 When the music stops, the guard spins around and everyone must freeze like statues.
- 4 Anyone caught moving goes back to the starting line.
- 5 The first one to tag the guard without being caught moving wins.

What it builds: Exercises body control, attention, and the ability to stop on cue. Playing in a group, kids practice taking turns and following rules while burning off energy.

Variations & tips: Have the statues strike funny poses or imitate animals, or play in pairs holding hands so they have to move in sync.

Safety: Check that the floor is clear so no one falls when stopping short.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

Veggie Stamp Art

Turn a potato or a beet into a stamp and fill the paper with colorful shapes.

PREP

MATERIALS



30-45

min



Materials

- A potato, a beet, or half a bell pepper
- Tempera paint in several colors
- A plate for the paint
- Sheets of paper
- A knife (for adult use only)
- An apron or clothes that can get messy

Prep ahead

Cover the table and put on aprons. An adult cuts the potato in half and, if they like, carves a simple shape into the surface. Pour a little tempera paint of each color onto the plate.

Step by step

- 1 Dip the flat side of the vegetable into the paint, without loading on too much.
- 2 Press the vegetable onto the paper, pushing down gently to leave a print.
- 3 Repeat with different colors and vegetables to create lots of shapes.
- 4 Combine the stamps into borders, flowers, or a free-form design.
- 5 Let the paper dry before hanging it up or giving it as a gift.

What it builds: Sparks creativity and fine motor skills, and shows that nature's shapes can be used to make art. Your child experiments with textures, colors, and repeating patterns.

Variations & tips: Compare the prints each vegetable leaves, or use a cut beet to tint the paper with its natural juice, no paint needed.

Safety: Only an adult uses the knife. Don't eat the vegetables once they've been painted.



We did it!

Date: _____



What happened? What was the most fun?

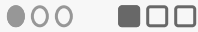
Draw what happened:

• • •

The Feelings Thermometer

Draw a special thermometer to measure how you're feeling and learn to cool down when it climbs too high.

PREP MATERIALS


 30-45
min


Materials

- A large sheet of paper or poster board
- Colored pencils
- A marker
- A clothespin or a cardboard arrow (optional)

Prep ahead

Find a quiet, comfortable spot to talk without rushing. Together, draw a tall, wide thermometer on the paper, with room for several levels of color.

Step by step

- 1 Color the thermometer from bottom to top: green for calm, yellow for restless, and red for big anger.
- 2 Talk about what the body feels at each color, like a racing heart or clenched hands.
- 3 Ask your child to point with the clothespin or their finger to the color they're at today.
- 4 Come up with an idea together for getting from red or yellow back down to green, like taking deep breaths or drinking water.
- 5 Hang the thermometer in their room to use at other times of day.

What it builds: Helps kids recognize and name emotions and notice how they feel in the body. Your child learns that getting angry is normal and that there are ways to come back to calm.

Variations & tips: Make a thermometer for the whole family, or add drawings of faces at each level for younger kids.

Safety: Keep the conversation calm and free of scolding; the goal is for your child to feel heard.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

Guess the Texture with Your Eyes Closed

Reach into a secret box and figure out what object you're touching using only your sense of touch.

PREP

MATERIALS



30-45

min



Materials

- A cardboard box with a lid
- A blindfold or scarf
- Objects with different textures: a sponge, a smooth stone, yarn, an avocado, a spoon, a leaf
- A notebook for writing things down

Prep ahead

An adult cuts a hand-sized hole in the box, or simply gets a cloth bag ready. Secretly gather several objects with very different textures without letting your child see them.

Step by step

- 1 Put the blindfold on your child, or ask them to keep their eyes shut tight.
- 2 Place an object inside the box and let them reach in to feel it.
- 3 Ask them to describe what they feel: whether it's soft, rough, cold, hard, or squishy.
- 4 Have them guess what the object is before pulling it out to look.
- 5 Write down in the notebook how many they guessed right, and talk about which one was the hardest.

What it builds: Sharpens the sense of touch and the vocabulary for describing textures. Your child focuses on a single sensation and practices figuring things out without using sight.

Variations & tips: Switch roles so your child hides the objects and the adults guess, or use only natural objects collected on a walk.

Safety: Don't use anything pointy or sharp. Check that everything going into the box is safe to touch.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

. . .

 EXPERIMENT

Ages 6–8

The Fireproof Balloon

Hold a flame near a balloon with water inside and discover why it doesn't pop like one filled with only air.

PREP MATERIALS



30-45

min



Materials

- Two balloons of the same size
- Tap water
- A candle
- Matches or a lighter (for adult use only)
- A bucket of water nearby, just in case
- A notebook for writing things down

Prep ahead

Do this activity outdoors or near a faucet, with a bucket of water on hand. Inflate one balloon with air only and tie it off. Fill the other balloon with tap water until it's about the size of a fist, then add a little air and tie it off. An adult lights and holds the candle.

Step by step

- 1 Ask your child which balloon they think will pop first when it's brought near the flame, and write down the prediction.
- 2 The adult carefully brings the air-only balloon near the flame, and you'll watch it burst right away.
- 3 Now the adult brings the water-filled part of the other balloon near the flame, without letting it touch the flame directly.
- 4 Notice that this balloon doesn't pop, even though the flame is heating it from below.
- 5 Write down what happened and compare the result with the prediction from the start.

What it builds: Teaches that water absorbs much of the flame's heat and keeps the balloon's rubber cooler, so it doesn't break. The air-only balloon has nothing to absorb the heat, which is why it pops so fast.

Variations & tips: Try more or less water inside the balloon to see how much it can take, or carefully touch the wet part afterward to feel that it's still just warm, not hot.

Safety: Only an adult uses the candle and matches. Keep hair and clothing away from the flame, and keep the bucket of water close by to put out any accident.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

. . .

 EXPERIMENT

Ages 6–8

The Toy Parachute

Build a parachute from a plastic bag and string, and discover how air slows a toy's fall.

PREP

MATERIALS



30-45
min



Materials

- A clean plastic bag
- Scissors
- String or yarn
- Tape
- A small, lightweight toy to be the parachutist
- A notebook for writing things down

Prep ahead

Cut a square about the size of a sheet of paper out of the bag. Cut four pieces of string of equal length. Tape one end of each string to each corner of the square, then gather the other ends and tie them to the toy. Find a safe, high spot, like a sturdy chair.

Step by step

- 1 Ask your child which will fall faster: the toy alone or the toy with the parachute, and write down the prediction.
- 2 Drop the toy by itself from the chair and count out loud how long it takes to reach the floor.
- 3 Now release the toy with the parachute open from the same height and count again.
- 4 Watch how the parachute fills with air and floats down more slowly than the toy alone.
- 5 Write down both times, compare them, and check whether the prediction from the start came true.

What it builds: Teaches that air pushes up against the parachute and slows the fall: that force is called air resistance. The bigger the canopy, the more air it catches and the more slowly the toy comes down.

Variations & tips: Try parachutes of different sizes, or cut a small hole in the center to see if it falls straighter. You can also compare a plastic bag with a sheet of paper.

Safety: Climb onto the chair carefully or ask an adult for help. Don't launch the toy at anyone's face, and keep plastic bags away from younger children.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:



Ages 9-12

Think big and work as a team

Ages 9–12

Think big and work as a team

On the doorstep of adolescence they crave autonomy, meaning and real challenges. They're motivated by building something useful, investigating, debating and collaborating. The activities in this chapter invite them to think critically, express more complex emotions and lead projects. Trust their abilities, hand them responsibility, and talk: at this age, quality time is also about listening.

26

ACTIVITIES

12

SCIENCE EXPERIMENTS

15

READY TO START NOW

In this chapter

● Marker Chromatography	45-60 min	● Extract DNA from a Strawberry	45-60 min
● DIY Water Clock	45-60 min	● DIY Electromagnet	45-60 min
● Secret Code Messages	45-60 min	● Vinegar and Baking Soda Rocket	45-60 min
● DIY Escape Room	45-60 min	● The Floating Egg and Density	45-60 min
● Invent Your Own Strategy Game	45-60 min	● Series and Parallel Circuits	45-60 min
● Balance and Challenge Course	45-60 min	● The Pendulum's Secret Rhythm	45-60 min
● Make and Learn to Juggle	45-60 min	● The Bridge That Holds the Most Weight	45-60 min
● Time Capsule	45-60 min	● Nature and Bird Journal	45-60 min
● The Big Conversations Jar	45-60 min	● Strategy Tournament with a Homemade Board	45-60 min
● Create Your Own Comic	45-60 min	● Design Your Visual Identity	45-60 min
● Flipbook: Screen-Free Animation	45-60 min	● Emotions Thermometer	45-60 min
● Your Flower Pot Garden	45-60 min	● DIY Layered Water Filter	45-60 min
● Red Cabbage pH Indicator	45-60 min	● The Yeast That Inflates a Balloon	45-60 min

EXPERIMENT

Ages 9–12

Marker Chromatography

Discover the hidden rainbow of colors lurking inside a black marker.

PREP

MATERIALS



45-60

min



Materials

- Coffee filters or absorbent paper towels
- Washable markers in several colors (especially black and brown)
- Glass cups or mugs
- Water
- Pencils or sticks to hang the paper from
- Scissors

Prep ahead

Cut the filter paper into strips about 1 inch wide and 5 inches long. Prepare one strip for each color you want to investigate. Fill the cups with about half an inch of water.

Step by step

- 1 With each marker, draw a thick dot about an inch from the bottom end of a strip. Note which color you used at the top in pencil.
- 2 Roll the top end of the strip around a pencil and rest it on the rim of the cup, so the bottom tip touches the water but the ink dot stays above it.
- 3 Watch how the water climbs up the paper and carries the ink along. Before it happens, predict which colors you think will appear.
- 4 Wait 10 to 20 minutes without moving the cup and watch how each ink separates into bands of different colors.
- 5 Take the strips out, let them dry, and compare them. Discuss why the black separated into so many colors while others did not.
- 6 Investigate as a family: why do some pigments climb higher than others? Write down your hypotheses.

What it builds: Introduces the scientific method (predict, observe, conclude) and the idea that many mixtures can be separated. Each pigment travels at a different speed depending on its size and how well it bonds with water, just like in real laboratories.

Variations & tips: Try ink from a non-permanent ballpoint pen, food coloring, or crushed flower petals. Compare markers from different brands to see which one has the most pigments.

Safety: Use the scissors carefully when cutting the paper. Use only washable markers, not permanent or solvent-based ones.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

 EXPERIMENT

Ages 9–12

DIY Water Clock

Measure time like the ancients did, using nothing but water and gravity.

PREP

MATERIALS



45-60
min



Materials

- Two identical plastic bottles (or one bottle and a clear cup)
- A thumbtack or a thin nail
- Tape
- Water
- Permanent marker
- A clock or stopwatch for calibrating

Prep ahead

Wash the bottles well and remove the labels. Ask an adult to help make the hole. Keep the clock handy to calibrate your invention at the end.

Step by step

- 1 Make a small hole in the center of one bottle's cap using the thumbtack. The smaller the hole, the slower the water will drip.
- 2 Fill that bottle with water and screw on the pierced cap. Place it upside down on top of the second, empty bottle and tape the necks together.
- 3 Flip it over so the water starts dripping into the bottom bottle. Watch the stream.
- 4 With the clock beside you, use the marker to mark the water level every minute on the bottle that's filling up.
- 5 Once you have your marks, empty it and repeat: now try to guess the time by looking only at your water clock, without the regular clock.
- 6 Reflect: does it always drip at the same speed? Investigate why it slows down when there is less water left on top.

What it builds: Explores water pressure and how people measured time before mechanical clocks. Calibrating your own marks builds precision, observation skills, and the idea that pressure drops as the water level goes down.

Variations & tips: Build several clocks with holes of different sizes and compare them. Challenge someone to guess when exactly 3 minutes have passed using only your clock.

Safety: Ask an adult for help piercing the cap with the thumbtack or nail, since the point can slip and cause a poke.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:



GAME

Ages 9–12

Secret Code Messages

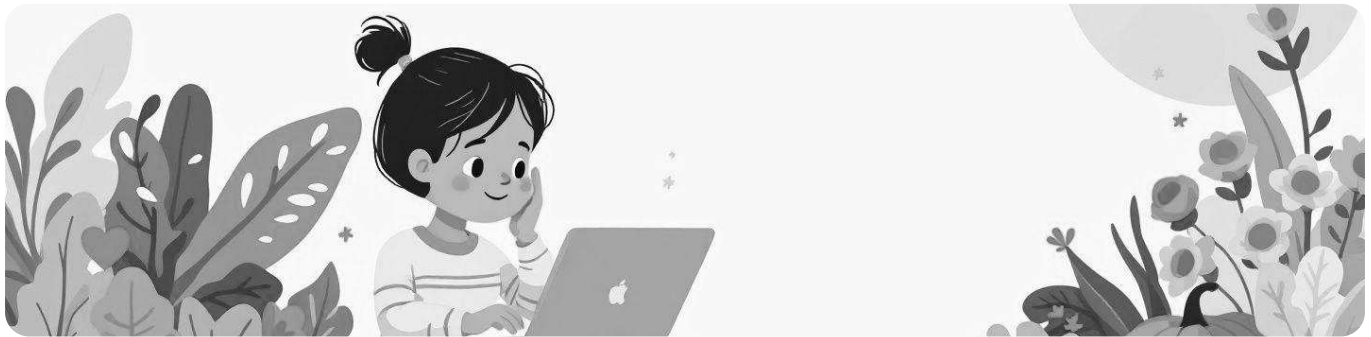
Invent your own cipher and send messages only your team can read.

PREP

MATERIALS



45-60
min



Materials

- Paper and pencil
- Ruler
- Cardboard to build a cipher wheel (optional)
- A brass fastener or thumbtack for the wheel (optional)
- Scissors (if you make the wheel)

Prep ahead

Gather everyone who is going to play. Decide which team will send messages and which will decode them. Get paper and pencil ready for everyone.

Step by step

- 1 Learn the Caesar cipher: write out the alphabet and, below it, another alphabet shifted by a fixed number of letters (for example, 3). A becomes D, B becomes E, and so on.
- 2 Each member writes a short message and encrypts it letter by letter using that shift. Agree on the key number in secret.
- 3 Swap the encrypted messages and try to crack them without knowing the key: test different shifts until the message makes sense.
- 4 Now invent your own code: you can swap letters for symbols, numbers, or drawings. Create a key table that only your team knows.
- 5 If you want, cut out two cardboard circles of different sizes with the alphabet around the edge and join them with a brass fastener so you can spin and encrypt faster.
- 6 Set up a final challenge: hide an encrypted clue somewhere in the house that leads to a small treasure.

What it builds: Introduces cryptography and logical thinking. Encrypting and decrypting builds reasoning, attention to detail, and the concepts of pattern and key, the foundation of modern computer security.

Variations & tips: Try encrypting by writing words backwards, or use a keyword that tells you how many positions to shift each letter. Time each other to see who decodes fastest.

Safety: If you build the cipher wheel, use the scissors carefully when cutting the cardboard.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

DIY Escape Room

Design a puzzle room at home and challenge your family to escape.

PREP

MATERIALS



45-60
min



Materials

- Paper and pencil
- Envelopes or boxes to hide clues in
- A combination lock if you have one (optional)
- Household objects to hide and use as clues
- A stopwatch or clock

Prep ahead

Choose who designs the room and who will solve it; you can swap roles afterward. The designer secretly plans the chain of clues in a safe room.

Step by step

- 1 Decide on a simple story: for example, you must find the key to open a chest before time runs out.
- 2 Design a chain of puzzles: each solved clue should lead to the next. Start from the end (the prize) and plan backwards.
- 3 Create a variety of puzzles: a math problem whose answer is a lock combination, a coded message, a hidden object, a riddle.
- 4 Hide the clues around the room inside envelopes, boxes, or under objects. Test the chain yourself from start to finish to make sure it works.
- 5 Welcome the players, explain the story and the rules, and start the timer (for example, 15 minutes).
- 6 Stay close and offer a hint if they get stuck. When it's over, talk about which puzzle was the hardest and why.

What it builds: Develops planning, logical thinking, and creativity. Designing for others forces you to anticipate how the players will reason, which builds empathy and project leadership.

Variations & tips: Raise the difficulty with clues that require several people to cooperate at once. Give it a theme: detectives, pirates, or a spaceship.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •



Ages 9–12

Invent Your Own Strategy Game

Design a board game from scratch with your own board, rules, and pieces.

PREP

MATERIALS



45-60

min



Materials

- Cardboard or poster board for the board
- Ruler and pencil
- Markers
- Bottle caps, coins, or buttons as game pieces
- A die (or a handmade paper die)
- Paper for writing down the rules

Prep ahead

Gather the materials on a large table. Decide as a family how many players the game will have; that will shape the design of the board.

Step by step

- 1 Agree on the goal of the game: reaching a finish line, conquering squares, collecting points, or capturing rival pieces.
- 2 Draw the board on the cardboard: a path of squares, a grid, or zones to conquer. Add special squares with effects.
- 3 Write the rules on a sheet of paper: how each piece moves, what happens on the special squares, and how you win.
- 4 Hand out the pieces and play a test round. Note which rules don't work or feel unfair.
- 5 Adjust the rules after testing: balance luck and strategy so that winning depends on thinking, not just the die.
- 6 Play the final version and give it a name. Explain it to someone else to check that the rules make sense.

What it builds: Builds strategic thinking, systems design, and negotiation. Balancing the rules teaches you to test, fail, and improve (iteration), and to write clear instructions for others.

Variations & tips: Add event cards that change the course of the game, or a cooperative mode where everyone plays against the board. Create a five-minute express version.

Safety: If you cut cardboard for the board or the pieces, use the scissors carefully.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

Balance and Challenge Course

Turn the yard or hallway into an obstacle course that puts your body to the test.

PREP

MATERIALS



45-60
min



Materials

- Rope or masking tape to mark lines
- Cushions, chairs, or boxes as obstacles
- Bottles or cans to weave around
- A stopwatch or clock
- Paper and pencil to record times

Prep ahead

Choose a wide, clear space, indoors or outside. Remove fragile objects from the route. Decide how many stations the course will have.

Step by step

- 1 Design the course together with several stations: a balance line made of rope, a zigzag between bottles, a jump over cushions, and a one-foot hopping stretch.
- 2 Mark the start and finish. Walk the course slowly once to check that it's safe and doable.
- 3 Time each participant's first attempt and write down the times.
- 4 Repeat several rounds trying to beat your own record. Talk about which station is hardest and why.
- 5 Add an extra challenge: run the course while balancing an object on your hand or head.
- 6 At the end, redesign one station together to make it harder and try again.

What it builds: Improves coordination, balance, and motor planning. Beating your own record teaches perseverance and self-improvement, and designing the course builds creativity and teamwork.

Variations & tips: Turn it into a team relay race, or add mental challenges between stations (a question or a math problem) to combine body and mind.

Safety: Clear the route of furniture with sharp corners and objects that could slip. Wear shoes with good grip and supervise the jumps to prevent falls.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

Make and Learn to Juggle

Craft your own juggling balls and master the art of the three-ball toss.

PREP

MATERIALS



45-60
min



Materials

- Sturdy balloons (3 per ball)
- Rice, lentils, or dried beans
- A cut plastic bottle or a paper funnel
- Scissors
- A spoon

Prep ahead

Gather the balloons and the filling on a table. Make a funnel by rolling up a sheet of paper or using the neck of a cut bottle.

Step by step

- 1 Fill a balloon with about three spoonfuls of rice using the funnel. Squeeze out the air and tie a tight knot.
- 2 Cut the neck off a second balloon and stretch it over the first one to cover the knot. Repeat with a third balloon turned the other way to reinforce it.
- 3 Make three matching balls of the same weight. Test them: they should fit in your palm and not bounce much.
- 4 Start by practicing with a single ball: toss it from one hand to the other in an arc at eye level.
- 5 Move on to two balls: toss the first, and when it reaches the top, toss the second underneath it. Practice the exchange without letting them collide.
- 6 Once you've mastered two, add the third. Practice in short bursts and celebrate every small win.

What it builds: Trains hand-to-hand coordination, visual tracking, and patience. Learning to juggle is a clear example of how steady practice improves a difficult skill, reinforcing a growth mindset.

Variations & tips: Invent tricks: tossing under your leg or behind your back. Practice in pairs, passing balls back and forth between two people.

Safety: Use the scissors carefully when cutting the balloons and the bottle. Practice away from fragile objects. Keep unused balloons out of reach of young children.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

Time Capsule

Save a message for your future self and open it a year from now.

PREP

MATERIALS



45-60
min



Materials

- A box with a lid or a glass jar with an airtight seal
- Paper and a ballpoint pen
- Printed photos, drawings, or small meaningful objects
- Tape and a marker

Prep ahead

Find a container that closes well and a quiet corner where you can write without rushing. Decide as a family when you will open the capsule: in six months, a year, or longer.

Step by step

- 1 Write a letter to your future self. Describe who you are now: what you like, your friends, your fears, and what you dream of achieving.
- 2 Answer a few questions to compare later: what's your favorite song? What do you want to be when you grow up? What makes you happy today?
- 3 Gather objects that capture this moment in your life: a photo, a ticket stub, a drawing, or a list of your favorite shows and books.
- 4 If you do it as a family, each person writes their own letter, plus one shared object that represents the group.
- 5 Put everything in the box, seal it well, and write on the outside the date before which it is forbidden to open it.
- 6 Hide the capsule in a safe place and mark the opening day on a calendar.

What it builds: Encourages introspection, emotional expression, and awareness of the passage of time. Comparing who you were with who you will become helps you appreciate your own growth and reflect on what really matters.

Variations & tips: Turn it into a yearly tradition: a new capsule every year. Or write crisscross letters, one for each family member, to be read together on opening day.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

The Big Conversations Jar

Questions for real talk, to discover what each person truly thinks and feels.

PREP

MATERIALS



45-60

min



Materials

- A jar, mug, or box
- Paper cut into strips
- Ballpoint pens for everyone
- A comfortable spot to sit in a circle

Prep ahead

Pick a moment with no rush and no screens, for example after dinner. Sit in a circle so everyone can see each other's faces.

Step by step

- 1 Each person secretly writes three or four questions on the strips of paper. They can be funny, curious, or deep.
- 2 Mix in some examples for inspiration: what moment are you proudest of? What would you change about the world? When did you feel brave? What would you like us to know about you?
- 3 Fold all the strips and drop them into the jar. Mix them well.
- 4 Taking turns, each person draws a question and answers it honestly. Afterward, the others can answer too if they want.
- 5 Agree on a golden rule: listen without interrupting and never judge an answer. Everything shared is respected.
- 6 When you finish, talk about which question made you think the most and save the jar to do it again another day.

What it builds: Develops emotional intelligence, active listening, and the ability to express complex ideas. Discussing and sharing feelings in a safe environment strengthens bonds and critical thinking.

Variations & tips: Create themed jars: one for memories, one for dilemmas to debate ("would you lie to help a friend?"), and one for future dreams. Invite grandparents or friends to join in.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

Create Your Own Comic

Invent a story with panels, speech bubbles, and a hero made just for you.

PREP

MATERIALS



45-60

min



Materials

- Sheets of paper
- Ruler and pencil
- Eraser
- Markers or colored pencils
- Black ballpoint pen for inking

Prep ahead

Gather the materials on a well-lit table. First think of a simple idea: a character, a problem they must solve, and an ending.

Step by step

- 1 Define your main character: what they're like, what they want, and their biggest challenge. Draw several faces until you find their expression.
- 2 Write the script in a few sentences divided into beginning, middle, and end. Decide how many panels you need to tell it.
- 3 Using the ruler, draw the panels on the page. Vary their size: important scenes can get a bigger panel.
- 4 Sketch each scene in pencil and add the speech bubbles and text boxes. Check that the reading order is clear.
- 5 Trace over the final lines with the black ballpoint pen, erase the pencil, and color it in. Add sound effects like 'BOOM!' or 'WHAM!' for extra punch.
- 6 Give it a title and a cover. Read it to your family and ask for their feedback to make the next story even better.

What it builds: Combines visual storytelling, writing, and drawing. Structuring a story in panels builds summarizing skills, time sequencing, and creativity, and communicating with image and text together develops visual language.

Variations & tips: Create a collaborative comic where each person draws one panel without fully seeing the previous one. Or turn a real family story into a comic.

Safety: If you cut the pages to bind the comic, use the scissors carefully.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

Flipbook: Screen-Free Animation

Bring your drawings to life with a pocket-sized movie that moves as you flip the pages.

PREP

MATERIALS



45-60
min



Materials

- A pad of sticky notes or lots of identical small sheets
- Pencil and eraser
- Black ballpoint pen
- A binder clip or paper clip to hold the sheets
- Scissors (if you cut the sheets)

Prep ahead

Get a thick pad of identical small sheets, or cut paper to the same size and clip it together along one side. Think of a simple movement to animate.

Step by step

- 1 Choose a simple animation: a bouncing ball, a growing flower, or a face that slowly breaks into a smile.
- 2 Start on the last page of the pad and draw the first moment of the movement, near the free edge.
- 3 On the page above it, repeat the drawing with just a tiny change. Use the previous page as a guide by looking through the paper against the light.
- 4 Keep going page by page with minimal changes. The more drawings you make and the less each one varies, the smoother the animation will be.
- 5 Trace over all the drawings with the black ballpoint pen so they show up well, and erase the pencil marks.
- 6 Hold the pad on one side and flip the pages with your thumb at a good speed. Your drawings come to life! Adjust anything that doesn't flow well.

What it builds: Shows how animation and persistence of vision work: many still images that, seen quickly, look like motion. It demands planning, persistence, and frame-by-frame attention to detail.

Variations & tips: Make longer flipbooks with complete stories, or combine two different animations. Try swapping them with friends to guess what movement each person drew.

Safety: If you cut the paper to make the sheets even, handle the scissors carefully.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

Your Flower Pot Garden

Grow your own herbs and watch them get bigger day by day.

PREP

MATERIALS



45-60
min



Materials

- A flower pot, a planter box, or recycled containers with holes in the bottom
- Potting soil
- Seeds or cuttings (basil, parsley, lentils, garlic, or an avocado pit)
- A watering can or a bottle with small holes in the cap
- A notebook and pencil for your growing journal
- Labels or sticks to mark each plant

Prep ahead

Choose a container with drainage holes in the bottom (or ask for help making them) and a spot with natural light by a window or on a balcony. Decide what you want to grow.

Step by step

- 1 Fill the container with soil almost to the rim, leaving an inch or two free. Gently press down the surface.
- 2 Plant the seeds at the depth the packet indicates (usually two or three times their size) or carefully bury the cuttings.
- 3 Water slowly until the soil is moist but not soggy. Place the flower pot where it will get light.
- 4 Mark each plant with a label and start a growing journal: write down the planting date and draw a picture of how it looks at the start.
- 5 Tend your garden every day: check the moisture with your finger and water only when the soil is dry. Record the changes in your journal.
- 6 When the first leaves sprout, measure their height every week and compare. If you planted herbs, use them in the family kitchen.

What it builds: Teaches responsibility, patience, and the life cycle of plants. Keeping a growing journal and measuring growth develops scientific observation and the steady care of a living thing under your watch.

Variations & tips: Run an experiment: plant two identical flower pots and leave one without light or without water to compare. Create a small shared garden where each family member cares for one plant.

Safety: If you need to make the drainage holes in the container, have an adult do it. Wash your hands after handling the soil.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

EXPERIMENT

Ages 9–12

Red Cabbage pH Indicator

Create a magic liquid that changes color to tell you whether something is acidic or basic.

PREP

MATERIALS



45-60
min



Materials

- Half a red cabbage
- Hot water (prepared by an adult)
- A strainer and a bucket or bowl
- Clear glass cups
- Spoons
- Substances to test: lemon juice, vinegar, baking soda, dish soap, soapy water, milk
- Notebook and ballpoint pen for recording

Prep ahead

An adult cuts the red cabbage into small pieces, puts them in a bucket, and adds hot water to cover them. After 15 minutes the water turns purple: that is your indicator. Strain the liquid into another container and let it cool. Set up a row of cups, one for each substance to test.

Step by step

- 1 Before you start, write your hypothesis: which substances do you think will be acidic and which basic? Write it in your notebook.
- 2 Pour a little red cabbage juice into each cup, in roughly equal amounts.
- 3 Add a different substance to each cup (a spoonful of lemon juice, of vinegar, of dissolved baking soda, and so on) and stir with a clean spoon.
- 4 Observe the colors: red or pink means acidic, green or yellow means basic, and purple means neutral. Record each color in a table.
- 5 Compare your results with your original hypothesis. Were you right? Mark the ones that surprised you.
- 6 Write a conclusion: order the substances from most acidic to most basic based on the color.

What it builds: Red cabbage juice contains pigments called anthocyanins that change color depending on pH. You practice the full scientific method: hypothesis, observation, recording data, and conclusion, just like in a real chemistry lab.

Variations & tips: Create a color scale on paper to classify new substances. Mix an acidic cup with a basic one and watch the color shift back toward purple. Try it with drinks like soda or juices.

Safety: An adult always handles the hot water. Do not taste or drink any of the mixtures. Wash your hands and the cups well when you finish, especially after using dish soap.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

. . .

 EXPERIMENT

Ages 9–12

Extract DNA from a Strawberry

Pull out with your own hands the strand of DNA that holds all of a strawberry's information.

PREP

MATERIALS



45-60
min



Materials

- 2 or 3 ripe strawberries
- A resealable or sturdy plastic bag
- Half a teaspoon of salt
- A tablespoon of dish soap
- Half a cup of water
- A fine strainer or cheesecloth
- A clear glass cup
- Ice-cold rubbing alcohol (provided by an adult, chilled in the refrigerator beforehand)
- A skewer stick

Prep ahead

Put the rubbing alcohol in the refrigerator at least 30 minutes ahead so it gets very cold.

Prepare the extraction solution by mixing the half cup of water, the salt, and the dish soap in a cup; stir gently so it doesn't make much foam.

Step by step

- 1 Put the strawberries in the bag, seal it, and mash them with your hands into a puree. Write your hypothesis: will you be able to see the DNA with the naked eye?
- 2 Add the water, salt, and dish soap solution to the bag and mix again, mashing gently for a minute.
- 3 Strain the mixture through the fine strainer or cheesecloth into the glass cup, keeping only the pink liquid.
- 4 Tilt the cup and have an adult slowly pour the cold alcohol down the side, forming a layer that floats on top of the liquid.
- 5 Watch the spot where the two layers meet: white, cloudy strands will appear. That is the DNA.
- 6 Carefully wind the strands around the skewer stick and lift them out. Note what you saw and write your conclusion.

What it builds: The dish soap breaks open the cell membranes, the salt helps the DNA clump together, and the cold alcohol makes the DNA (which doesn't dissolve in alcohol) become visible. You discover that all the information of a living thing fits in strands so tiny we normally can't see them.

Variations & tips: Repeat with other soft fruits like banana or kiwi and compare which one gives the most DNA. Try room-temperature alcohol and record whether it works as well as the cold kind.

Safety: An adult always handles the alcohol and keeps it away from flames. Do not taste any mixture. Wash your hands when you finish.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

 EXPERIMENT

Ages 9–12

DIY Electromagnet

Turn a nail into a magnet you can switch on and off with a battery.

PREP

MATERIALS



45-60
min



Materials

- A large iron nail
- Thin enameled copper wire (three feet or more)
- A large D battery or a 4.5V lantern battery
- Electrical tape
- Metal paper clips, pins, or tacks
- A small piece of sandpaper or a knife (used by an adult)
- Notebook and ballpoint pen for recording

Prep ahead

An adult scrapes both ends of the copper wire with the sandpaper to remove the enamel and expose the bare metal, so it can make contact with the battery. Gather several paper clips or pins for the tests.

Step by step

- 1 Write your hypothesis: how many paper clips do you think the nail will be able to lift? Write it in your notebook.
- 2 Wind the wire around the nail in tight, neat coils, leaving both ends free. Count and record how many turns you made.
- 3 Connect one end of the wire to each terminal of the battery and secure them with electrical tape.
- 4 Bring the tip of the nail close to the paper clips and observe: now the nail attracts them. Count how many it can lift and record it.
- 5 Disconnect one end from the battery and watch the paper clips fall off: the magnetism disappears.
- 6 Change one variable: add more turns of wire and repeat the test. Compare the results and write your conclusion.

What it builds: When electric current flows through the coiled wire, it creates a magnetic field that turns the nail into a magnet. You discover the link between electricity and magnetism, and that more turns make the electromagnet stronger. It's the same principle behind motors and doorbells.

Variations & tips: Compare two nails with different numbers of turns. Try nails of different sizes or a more powerful battery and record the changes in strength.

Safety: Use only batteries, never a wall outlet. The wire and battery can heat up if left connected too long, so disconnect them between tests. An adult does the wire scraping.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

 EXPERIMENT

Ages 9–12

Vinegar and Baking Soda Rocket

Launch a rocket into the air using only a chemical reaction and a lot of pressure.

PREP

MATERIALS



45-60
min



Materials

- A small plastic bottle with a press-on cap (like a medicine bottle or a sports drink bottle)
- A cork that fits snugly in the bottle's mouth
- Vinegar
- Baking soda
- A piece of paper towel
- Three pencils to use as legs (with tape)
- Tape
- Notebook and ballpoint pen for recording

Prep ahead

This activity is done outdoors, in a yard or open space, away from people, windows, and cars. Build the rocket by taping three pencils to the sides of the bottle so it can stand upside down, with the cork facing the ground. Check that the cork seals the mouth with a little pressure.

Step by step

- 1 Write your hypothesis: how high do you think the rocket will go, and how many seconds until it launches?
- 2 Put some vinegar in the bottle, filling about a third of it.
- 3 Wrap two spoonfuls of baking soda in a piece of paper towel, forming a packet that fits through the mouth.
- 4 Drop the packet into the bottle, cap it quickly with the cork, set the rocket upside down on its legs, and step several paces back.
- 5 Watch the reaction: gas forms, pressure builds, and the cork shoots out, launching the rocket. Time how long it took.
- 6 Repeat, changing one variable (more baking soda or more vinegar), measure again, and write your conclusion about which amount worked best.

What it builds: The vinegar and baking soda react and produce carbon dioxide. The gas raises the pressure inside the bottle until the cork gives way and pushes the rocket upward. It's an example of the principle of action and reaction, the same one that powers real rockets.

Variations & tips: Decorate your rocket with cardboard fins and record whether it flies straighter. Compare how high it goes with different ingredient ratios and make a simple graph of your measurements.

Safety: Always do it outdoors and with an adult. Never point the cork toward people, animals, or windows. Step back as soon as you put the cork in, because it can shoot out with force. Wear safety goggles if you have them.

**We did it!**

Date: _____



What happened? What was the most fun?

Draw what happened:

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 EXPERIMENT

Ages 9–12

The Floating Egg and Density

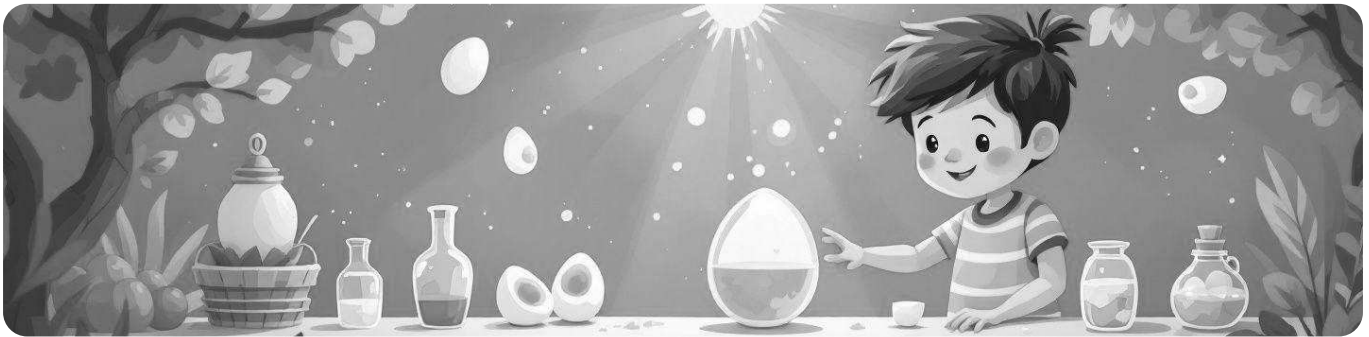
Make an egg float in water without touching it, just by changing the liquid.

PREP

MATERIALS

●○○○

■□□□


45-60
min


Materials

- A raw egg
- Two tall clear glass cups
- Salt
- Water
- A spoon
- Food coloring (optional)
- Notebook and ballpoint pen for recording

Prep ahead

Fill both cups halfway with water. Leave one with plain water and have the salt ready to prepare the second. Work on a surface that's easy to clean.

Step by step

- 1 Write your hypothesis: what will happen if you put an egg in plain water, and what will happen in water with a lot of salt?
- 2 Carefully place the egg in the cup of plain water and observe: it sinks to the bottom. Note it down.
- 3 In the other cup, add salt little by little and stir until a good amount dissolves, counting how many spoonfuls you use.
- 4 Place the egg in the salt water and watch how it now floats. Record how many spoonfuls of salt you needed.
- 5 Extra experiment: with the egg floating, slowly add plain water down the side of the cup to form a layer, and watch the egg float in the middle.
- 6 Write your conclusion explaining why the egg floats in salt water but not in plain water.

What it builds: Salt water is denser than plain water, so it pushes the egg upward with more force. You learn the concept of density and why it's easier to float in the ocean than in a pool. You practice measuring and recording variables.

Variations & tips: Try sugar instead of salt and compare how many spoonfuls you need. Try to form several layers of liquids with different densities (salt water, plain water, oil) and watch how they arrange themselves.

Safety: The egg stays raw, so wash your hands and surfaces well when you finish and don't leave it at room temperature. Do not drink the salt water.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

 EXPERIMENT

Ages 9–12

Series and Parallel Circuits

Build your own circuits and discover why some light bulbs shine brighter than others.

PREP

MATERIALS



45-60
min



Materials

- Two or three small flashlight bulbs with sockets (or low-voltage LED bulbs)
- A large battery or a 4.5V lantern battery
- Wires with clips or copper wire
- Electrical tape
- A homemade switch (a paper clip and two tacks on cardboard) optional
- Notebook and ballpoint pen for recording

Prep ahead

Prepare the wires by stripping the ends bare so they can make contact. If you use plain wire, an adult scrapes the ends. Gather the bulbs in their sockets and keep the battery handy.

Step by step

- 1 Write your hypothesis: if you connect two bulbs in a row, will they shine the same, brighter, or dimmer than a single one?
- 2 Build a series circuit: connect the battery, one bulb, then the other, and back to the battery, all in a single line. Observe the brightness and note it down.
- 3 Disconnect one bulb from the series circuit and watch what happens to the other. Record the result.
- 4 Now build a parallel circuit: each bulb connected separately to the two terminals of the battery. Compare the brightness with the previous case.
- 5 Disconnect one bulb from the parallel circuit and see if the other stays lit. Note the difference from the series circuit.
- 6 Write your conclusion: explain in which circuit the bulbs shine brighter and in which one a bulb stays lit even if the other fails.

What it builds: In a series circuit the current passes through all the bulbs in a row, so if one fails they all go out and they shine dimmer. In parallel each bulb has its own path, they shine brighter and work independently. That's why houses use parallel circuits.

Variations & tips: Add a homemade switch with a paper clip to turn the circuit on and off. Add a third bulb and predict how the brightness will change in each type of circuit before testing it.

Safety: Use only batteries, never a wall outlet. If the wires or battery get hot, disconnect them. An adult does the wire scraping.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

 EXPERIMENT

Ages 9–12

The Pendulum's Secret Rhythm

Discover what makes a pendulum swing faster or slower, like a physics detective.

PREP

MATERIALS

● ○ ○

■ □ □


45-60
min


Materials

- Sturdy string or yarn
- A nut, an old key, or a small heavy object
- A ruler or measuring tape
- A clock or stopwatch (an adult's is fine)
- A chair or the edge of a table to hang the pendulum from
- Tape
- Notebook and ballpoint pen for recording

Prep ahead

Find a spot where you can hang a string and the weight can swing freely without hitting anything. Tie the heavy object to one end of the string. Prepare a table in your notebook with columns for the string length and the swing time.

Step by step

- 1 Write your hypothesis: what will make the pendulum go faster, a shorter string or a heavier weight?
- 2 Hang the pendulum with a known string length (for example 12 inches) and measure it with the ruler.
- 3 Release the weight from one side and count how many complete swings it makes in 15 seconds. Note it down.
- 4 Change only the string length (make it shorter or longer) and repeat the swing count in 15 seconds.
- 5 Now keep the same length but change the weight, and count again. Record everything in your table.
- 6 Compare your data and write your conclusion: what changed the pendulum's rhythm, the length or the weight?

What it builds: The time a pendulum takes to swing back and forth depends on the length of the string, not the weight. A shorter string swings faster. You learn to isolate variables, measure carefully, and draw conclusions from data, just like Galileo when he studied pendulums.

Variations & tips: Build two pendulums of different lengths and watch them swing together to see how they fall out of sync. Use your pendulum as a stop-watch: adjust the string so each swing lasts about one second.

Safety: Make sure the pendulum won't hit anyone or fragile objects as it swings. Hang the string from a firm support.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

 EXPERIMENT

Ages 9–12

The Bridge That Holds the Most Weight

Put on your engineer's hard hat and build a paper bridge that holds as much weight as possible.

PREP

MATERIALS



45-60
min



Materials

- Several sheets of paper or poster board
- Tape
- Two books or boxes of the same height to use as supports
- A cup or small bucket to hang weight from
- Coins, nuts, or pebbles to add as weight
- A ruler
- Notebook and ballpoint pen for recording

Prep ahead

Separate the two books or boxes by a fixed distance (for example 8 inches) so your bridge always spans the same gap. Prepare a table in your notebook to note the bridge design and how much weight it held.

Step by step

- 1 Write your hypothesis: which paper shape do you think will hold the most weight, a flat sheet, one folded like an accordion, or a tube?
- 2 Build a first bridge by resting a flat sheet between the two books and test how much weight it holds before it gives way. Note it down.
- 3 Design a second bridge by folding the paper into an accordion or rolling it into tubes, and span it between the supports.
- 4 Add weight little by little on the center of the bridge and count how much it holds before it bends or falls. Record the result.
- 5 Test at least one more design, changing only one thing at a time (more folds, more layers, columns underneath).
- 6 Compare all the designs and write your conclusion about which structure was the strongest and why.

What it builds: The shape of a material changes its strength a lot: folded or rolled paper holds much more than flat because it distributes forces better. You practice engineering design: test, measure, improve, and test again, comparing one variable at a time.

Variations & tips: Hold a team competition to see whose bridge holds the most with the same amount of paper. Research real bridge shapes (arches, triangles) and try to copy them with your paper.

Safety: Use the tape and scissors carefully. Stack the weight steadily so it doesn't fall on your feet.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

Nature and Bird Journal

Become an explorer and record the birds and plants in your neighborhood like a field scientist.

PREP

MATERIALS



45-60

min



Materials

- A notebook for your field journal
- Ballpoint pen and colored pencils
- Binoculars if you have them (optional)
- A magnifying glass
- A watch to note the time
- Comfortable clothes and water

Prep ahead

Choose an outdoor spot to observe: a park, a plaza, a yard, or the walk to school. Set up your journal by writing the date, time, and place on the first page. If you go somewhere new, always go with an adult.

Step by step

- 1 Before heading out, write a question to investigate: for example, which birds show up earliest in the morning?
- 2 Find a quiet spot, stay silent for a few minutes, and calmly observe your surroundings.
- 3 Note and draw the birds you see: their size, colors, what they were doing, and at what time. Use the magnifying glass to look at leaves, insects, or seeds.
- 4 Also record the weather, the plants, and any animal tracks you find.
- 5 If you can, return to the same spot another day at the same time and compare what changed.
- 6 At the end, review your records and write a conclusion that answers your original question.

What it builds: You learn to observe closely, record data in the field, and compare observations over time, which is what biologists do. You also get to know the nature of your own neighborhood better and develop patience and respect for living things.

Variations & tips: Make a list of the birds you manage to identify and name them with help from a book or an adult. With your team, compare your journals and make a map of where you saw each species.

Safety: Always go out with an adult and don't wander off alone. Watch the animals from a distance, without touching or chasing them. Don't eat any plants or berries you find.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

. . .



Ages 9–12

Strategy Tournament with a Homemade Board

Design and play a strategy tournament where the best thinker wins.

PREP

MATERIALS



45-60
min



Materials

- Poster board or cardboard for the board
- Markers and ruler
- Game pieces (buttons, caps, or pebbles in two colors)
- Paper and ballpoint pen to keep score
- A die (optional)

Prep ahead

Gather two or more players. On the poster board, draw a grid board (for example 8 by 8) or the board for whatever game you like. Prepare the pieces in two colors and a sheet to record the tournament results.

Step by step

- 1 Choose a strategy game for the board, like an expanded tic-tac-toe, checkers, or a capture game you invent yourselves.
- 2 Write the rules clearly together so everyone plays the same way, including how to win.
- 3 Play a first test round and adjust the rules if anything feels confusing or unbalanced.
- 4 Set up a tournament: note who plays whom and keep track of the scores on the sheet.
- 5 After each game, talk about which strategy worked best and what you would do differently next time.
- 6 At the end of the tournament, add up the points, reward the winner, and propose an improvement to the rules for next time.

What it builds: Strategy games train logical thinking, planning, and the ability to anticipate the other player's moves. By creating and adjusting the rules as a team, you also practice negotiating, being fair, and learning from every game.

Variations & tips: Invent a completely new game with your own pieces and rules. Add a die for a bit of chance and watch how the strategy changes. Organize a championship over several days.

Safety: Remember that what matters is having fun and respecting your opponent, win or lose. If someone gets upset, take a break and talk it out before continuing.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

Design Your Visual Identity

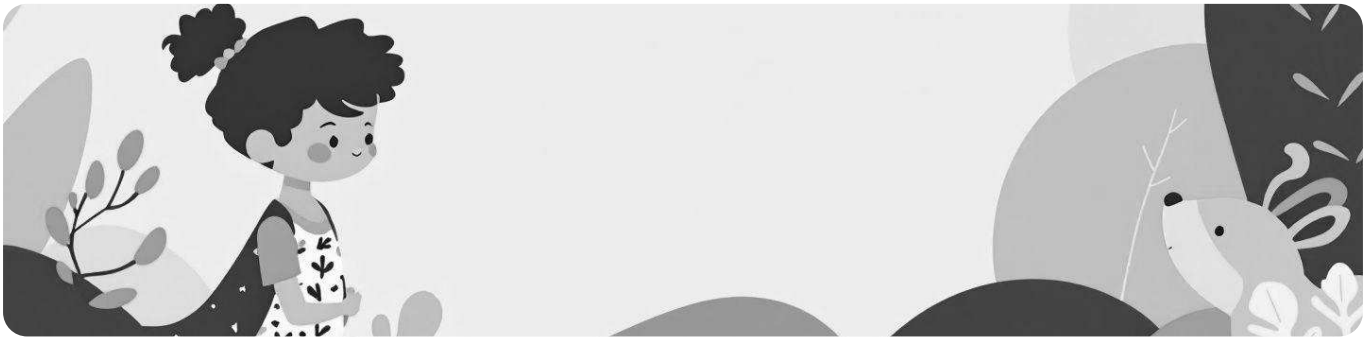
Create your own logo, colors, and typeface like a real brand designer.

PREP

MATERIALS



45-60
min



Materials

- White paper and scratch paper
- Colored pencils, markers, and a ballpoint pen
- A ruler
- Scissors and glue
- Magazines or colored paper to cut from (optional)

Prep ahead

Think of something you want to create an identity for: your name, a club you invent with your friends, a band, a team, or a project. Gather the materials on a well-lit table.

Step by step

- 1 Brainstorm: write down words and draw symbols that represent what you want to convey (cheerful, sporty, mysterious, eco-friendly).
- 2 Choose a palette of three or four colors that go together and test how they look side by side on a draft.
- 3 Design your logo by making several different sketches before deciding; combine a simple image with your name or initials.
- 4 Choose a lettering style by drawing it by hand and check that it reads well both large and small.
- 5 Make a clean copy of your final design on a sheet of paper, minding the colors and the details.
- 6 Apply your identity to something real: an ID card, a poster, a notebook cover, or a badge for your team.

What it builds: You discover how design communicates ideas and emotions through shapes, colors, and typography. You practice a designer's creative process: exploring many options, choosing, and refining, instead of settling for the first idea.

Variations & tips: As a team, design the identity of a shared club, then each person applies the logo to a different object. Make a black-and-white version of your logo and check that it's still recognizable.

Safety: Use the scissors carefully when cutting.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

Emotions Thermometer

Build a thermometer to measure and understand what you feel, just like you measure temperature.

PREP

MATERIALS



45-60

min



Materials

- Poster board or cardboard
- Colored markers
- A ruler
- Scissors
- A piece of yarn or a paper strip for the movable pointer
- A notebook for tracking emotions
- Ballpoint pen

Prep ahead

Find a quiet spot. First talk with an adult about the idea that all emotions are normal and serve a purpose. Get the poster board and markers ready to build the thermometer.

Step by step

- 1 On the poster board, draw a large vertical thermometer divided into levels from 1 (very calm) to 10 (very intense emotion).
- 2 Color the levels with shades that go from cool to warm and write example emotions next to each zone (calm, happy, nervous, angry).
- 3 Cut out an arrow or strip that you can move along the thermometer to mark how you feel.
- 4 Each day for a week, mark what level you're at during different moments and note it in your notebook with the date and what happened.
- 5 When you reach a high level, try a strategy to lower the intensity: take a deep breath, drink water, or walk around a bit, and record whether it worked.
- 6 At the end of the week, review your records and talk with an adult: which situations raise your thermometer and what helps you bring it down?

What it builds: You learn to recognize and name your emotions and to notice their intensity before you react. Keeping a record helps you spot patterns and choose strategies to calm down when you get angry or very nervous. It's the foundation of emotional intelligence.

Variations & tips: Make a family or team thermometer where everyone marks how they feel when they get home. Create a personal list of things that help you come down a level and keep it handy.

Safety: Remember that there are no bad emotions and that you can always ask a trusted adult for help if a feeling feels too strong or hard to handle.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:



EXPERIMENT

Ages 9–12

DIY Layered Water Filter

Build a filter with layers of stones and sand and watch dirty water come out much cleaner.

PREP

MATERIALS



45-60
min



Materials

- A large empty plastic bottle (an adult cuts off the base)
- A piece of cotton or an old kitchen cloth
- Clean sand
- Small stones or gravel
- Ground charcoal (the grilling kind, prepared by an adult)
- A strainer for washing the sand and stones
- A clear glass jar or cup to catch the water
- Dirty water made with dirt, leaves, and a little oil
- Notebook and ballpoint pen for recording

Prep ahead

An adult cuts off the base of the bottle so it works like a funnel and can stand upside down on the glass jar. Wash the sand and stones well with the strainer under the tap until the water runs clear. In a bucket, prepare the dirty water by mixing water with a little dirt, leaves, and a few drops of oil.

Step by step

- 1 Write your hypothesis: will the water come out totally clean, a little clearer, or just as dirty?
- 2 Put the bottle upside down on the jar and place the cotton or cloth at the bottom first, next to the mouth.
- 3 Add a layer of ground charcoal, then a layer of sand on top, and a layer of small stones on top of that.
- 4 Observe and draw your assembled filter: note the order of the layers, from the finest at the bottom to the coarsest at the top.
- 5 Slowly pour the dirty water in from the top and watch it drip down through each layer.
- 6 Compare the water that falls into the jar with the water from the bucket and note what changed in the color and in the floating bits.
- 7 Write your conclusion: which layer do you think trapped the most dirt, and why did your hypothesis hold up or not?

What it builds: Each layer traps particles of a different size: the stones stop the biggest bits, the sand holds the fine ones, and the charcoal helps remove odors and some impurities. This is similar to how water is cleaned at a treatment plant, separating what doesn't dissolve. You learn that filtering improves the water's appearance but doesn't make it safe to drink.

Variations & tips: Change the order of the layers or remove one and record whether the water comes out just as clear. Try more or less charcoal and compare the results by drawing each attempt.

Safety: Never drink the filtered water, even if it looks clean, because it may still have germs. An adult cuts the bottle and prepares the charcoal. Wash your hands when you finish.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •



EXPERIMENT

Ages 9–12

The Yeast That Inflates a Balloon

Wake up sleeping yeast with sugar and watch it inflate a balloon all on its own.

PREP

MATERIALS



45-60
min



Materials

- A packet of dry yeast (the bread-making kind)
- Three identical small plastic bottles
- Three balloons
- Sugar
- Warm water
- Cold water
- A measuring spoon
- A clock or stopwatch
- Notebook and ballpoint pen for recording

Prep ahead

Prepare warm water from the tap, not so hot that it burns, and keep cold water on hand too. Stretch each balloon a bit before starting so it inflates more easily. Number the three bottles with a marker: bottle 1 will have sugar and warm water, bottle 2 no sugar, and bottle 3 sugar but with cold water.

Step by step

- 1 Write your hypothesis: in which bottle do you think the balloon will inflate the most, and why?
- 2 In bottle 1, put one spoonful of yeast, two spoonfuls of sugar, and warm water up to a quarter of the bottle; swirl gently.
- 3 In bottle 2, put only yeast and warm water, no sugar. In bottle 3, put yeast, sugar, and cold water.
- 4 Place a balloon over the mouth of each bottle, covering it well so no gas escapes.
- 5 Observe for a few minutes: the yeast eats the sugar and releases carbon dioxide, which inflates the balloon.
- 6 Every five minutes, measure the size of each balloon with a measuring tape and record the data in a table.
- 7 Compare the three bottles and write your conclusion about what the yeast needs to produce more gas.

What it builds: Yeast is a tiny living fungus. When it has sugar and warmth it wakes up and, as it feeds, releases carbon dioxide in a process called fermentation. That same gas is what forms the bubbles that make bread fluffy. You learn that a microscopic living thing can do work we can see with the naked eye.

Variations & tips: Try honey or flour instead of sugar and record whether the yeast produces more or less gas. Repeat by leaving one bottle in a warmer spot and another in the refrigerator to compare the effect of temperature.

Safety: Do not taste any mixture. An adult prepares the warm water so it doesn't burn. If a bottle ends up closed with a lot of gas, open it carefully over the sink. Wash your hands when you finish.



We did it!

Date: _____



What happened? What was the most fun?

Draw what happened:

• • •

Activities by type

Experiment

- Sink or Float? (0-2 years)
- Magic Shadows (0-2 years)
- Disappearing Ice (0-2 years)
- Ball Ramp (0-2 years)
- Catch the Light (0-2 years)
- Falling Paper (0-2 years)
- Bubble Bottle (0-2 years)
- Rolling on Every Surface (0-2 years)
- The Baking Soda Volcano (3-5 years)
- The Dancing Colors (3-5 years)
- Oil and Water Won't Mix (3-5 years)
- White Flowers That Change Color (3-5 years)
- The Candle Under the Glass (3-5 years)
- The Balloon That Blows Itself Up (3-5 years)
- Seeds Sprouting in Cotton (3-5 years)
- The Magnet and Its Hidden Treasures (3-5 years)
- A Rainbow from Water and a Mirror (3-5 years)
- Magic Rainbow Milk (3-5 years)
- The Pepper That Runs Away (3-5 years)
- The Magic Tower of Liquids (6-8 years)
- Balloon Rocket on a String (6-8 years)
- The Magic Electric Balloon (6-8 years)
- Salt Crystals That Grow on Their Own (6-8 years)
- Homemade Lava Lamp (6-8 years)
- Secret Messages in Invisible Ink (6-8 years)
- The Goo That's Liquid and Solid at Once (6-8 years)
- The Hidden Colors Inside Markers (6-8 years)
- The Backyard Sundial (6-8 years)
- The Paper Bridge That Holds Weight (6-8 years)
- The Fireproof Balloon (6-8 years)
- The Toy Parachute (6-8 years)
- Marker Chromatography (9-12 years)
- DIY Water Clock (9-12 years)
- Red Cabbage pH Indicator (9-12 years)
- Extract DNA from a Strawberry (9-12 years)
- DIY Electromagnet (9-12 years)
- Vinegar and Baking Soda Rocket (9-12 years)
- The Floating Egg and Density (9-12 years)
- Series and Parallel Circuits (9-12 years)
- The Pendulum's Secret Rhythm (9-12 years)
- The Bridge That Holds the Most Weight (9-12 years)
- DIY Layered Water Filter (9-12 years)
- The Yeast That Inflates a Balloon (9-12 years)

Sensory

- Treasure Basket (0-2 years)
- Bubble Chasers (0-2 years)
- Calm-Down Bottle (0-2 years)
- Splash and Pour (0-2 years)
- Texture Blanket (0-2 years)
- Homemade Squishy Dough (3-5 years)
- The Mystery Box (3-5 years)
- Fluffy Foam Painting (3-5 years)
- The Mystery Touch Box (6-8 years)
- Guess the Texture with Your Eyes Closed (6-8 years)

Active play

- Little Kitchen Band (0-2 years)
- Pillow Obstacle Course (0-2 years)
- Scarf Dance (0-2 years)
- The Pillow Obstacle Course (3-5 years)

- Bottle Bowling (3-5 years)
- Homemade Obstacle Course (6-8 years)
- Homemade Bottle Bowling (6-8 years)

- Balance and Challenge Course (9-12 years)
- Make and Learn to Juggle (9-12 years)

Game

- Tumbling Towers (0-2 years)
- In and Out (0-2 years)
- Roll It Back and Forth (0-2 years)
- The Pretend Store (3-5 years)
- The Color Hunt (3-5 years)
- The At-Home Treasure Hunt (3-5 years)
- Invent Your Own Board Game (6-8 years)

- Treasure Hunt with Clues (6-8 years)
- The Freeze-Statue Race (6-8 years)
- Secret Code Messages (9-12 years)
- DIY Escape Room (9-12 years)
- Invent Your Own Strategy Game (9-12 years)
- Strategy Tournament with a Homemade Board (9-12 years)

Connection

- Where's Mommy? Peekaboo! (0-2 years)
- Faces in the Mirror (0-2 years)
- Sing-Along Massage (0-2 years)
- The Faces of Feelings (3-5 years)
- The Calm-Down Jar (3-5 years)

- The Calm-Down Jar (6-8 years)
- The Feelings Thermometer (6-8 years)
- Time Capsule (9-12 years)
- The Big Conversations Jar (9-12 years)
- Emotions Thermometer (9-12 years)

Arts & crafts

- Yogurt Finger Paint (0-2 years)
- Sponge Stamps (0-2 years)
- Potato and Veggie Stamps (3-5 years)
- Leaf and Flower Collage (3-5 years)
- Bubble-Print Paintings (3-5 years)
- Rainbow Salt Mandala (6-8 years)

- Cardboard Puppets and a Puppet Show (6-8 years)
- Veggie Stamp Art (6-8 years)
- Create Your Own Comic (9-12 years)
- Flipbook: Screen-Free Animation (9-12 years)
- Design Your Visual Identity (9-12 years)

Nature

- Backyard Treasures (0-2 years)
- Plant a Seed (0-2 years)
- Backyard Bug Safari (3-5 years)
- Nature Treasure Walk (3-5 years)

- Nature Explorer's Journal (6-8 years)
- The Plant That Chases the Light (6-8 years)
- Your Flower Pot Garden (9-12 years)
- Nature and Bird Journal (9-12 years)

One last thought

This isn't about eliminating screens completely — it's about balance. Every moment your kids play, explore or laugh with you is a seed. Thank you for choosing to be present. Those are the moments they will truly remember.